Functional Foods: Concept, Technology and Health Benefits

Functional Foods: Concept, Technology and Health Benefits

CENTER FOR DEVELOPMENT OF TECHNICAL EDUCATION, IIT KANPUR AND COMMONWEALTH OF LEARNING (COL)

COMMONWEALTH OF LEARNING (COL)

Contents

Course Description

Due to increased cost of health-care and lifestyle related diseases, the consumers are shifting towards the functional foods for health promotion and disease prevention. Functional foods are prepared to promote health by targeting a physiological process that may help in disease prevention. With the expansion of food industry, the use of functional foods in healthcare is increasing day by day and consumers are now health-conscious & preferring functional foods as a first line of defense against diseases rather than the pharmaceuticals. So the consumers are changing their mindset towards "Prevention is better than cure". Many functional compounds have been reported for prevention and curing of these diseases. Carotenoids, lutein, lycopene, soluble and insoluble dietary fibers, plant sterols, conjugated lenoleic acid (CLA), β-lactoglobulin, lactoferrin are few of the examples of functional food ingredients. Other functional food components like probiotics, prebiotics and synbiotics are also important. Various probiotics appear to induce anti-inflammatory responses which inhibit carcinogenesis and also improve the biological defense mechanisms.

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Course Content

- Basic Concepts of Functional Foods and Nutraceuticals
- Functional Foods: Categorization and Sources
- Probiotics: As Functional Foods
- Role of Functional Foods in Health and Disease

Course Audience

- Students enrolled in UG and PG in Food Science/Food Science and Nutrition Food Technology / Food Engineering, Biotechnology, Clinical Nutrition and Dietetics, Applied Nutrition, Post Harvest Technology, Biochemical Engineering, Livestock Products Technology and Life Sciences
- Faculty of SAUs
- Food Scientists
- Nutritionist and Entrepreneurs
- Health Professionals
- Consumers/ General Public

Outcomes of this Course

- Regulatory Systems for Functional Foods
- Functional Food Market in India & Global Market
- Prebiotics, Synbiotics, Types of Prebiotics and Their Mechanism of Action
- Role of Functional Foods in Management of Diabetes, Parkinson Disease (PD), Cancer & CVDs

Certificates

Qualifying students will be given certificates based on their involvement and performance. Participation certificate and Competency certificate will be issued by Center for Development of Technical Education, IIT Kanpur and Commonwealth of Learning (COL), Canada.

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Transcript

Welcome to all of you. This is the 1st lecture of the 1st week. And in this week we will go through the definitions of Functional Foods and Nutraceuticals; their origin, various sources of functional foods. What is the significance of the functional foods, their regulatory aspects and many more things. First we discuss the definitions of the functional foods. There are many definitions provided by different peoples to define functional foods. But here we mention the very reliable definition that is given by Institute of Food Technologists. They define food and food components that provide health benefit beyond the basic nutrition. Means, they are playing the basic role; as well as beyond that basic role also including the conventional foods, fortified, enriched or enhanced food and dietary supplements. They provide essential nutrients often beyond the quantities that are required for the normal maintenance, growth and development of other biological active components; that impart

health benefits or desirable physiological
benefits.

Another definition given by Institute of Food and Information Council states that functional foods provide health benefits beyond basic nutrition. American Institute of Dietetics; they define functional foods are those foods that have health benefits beyond the nutrients they contain. We can see that in all the three definitions; one thing is common, that they are providing the basic as well as the advanced nutrition also. From where these concept of functional foods originates. The concept of functional foods originates from the ancient China and then shifted to Japan, long ago. Arai in 2005 gave the concept of medicines and foods are isogonics. Similarly the father of medicine, Hippocrates; he defines, he gives the statement that, "Let food be the medicine and medicine be the food". Hassler in 2001 renew the same term of functional foods. Japan Ministry of Health and Welfare initiated the world's first policy of legally permitted, the commercialization of selected functional foods named 'Food for specified Health Use', that is FOSHU.

Now here we can see that; we know that food provides or consists of carbohydrates, fats, protein, vitamin, mineral, water and roughage. And these all components, they are very important for the normal body function; like for the growth of the human beings, for the repairing of the tissues, for the energy production as well as the protection. Then what happened if we are adding some bioactive compounds in this food, then it becomes functional. Similarly we can add also and here if we remove some particular harmful ingredients or deleterious compound that is present in food, then also it is called functional foods. And the these functional they foods provide the basic nutrition with the physiological health benefits, and also reduce the risk of chronic diseases.

A natural food product to make it functional, there are so many ways. We can add something to it that is beneficial to us. We can remove something that is harmful. Even we can enhance something from the outer sources to make it functional. And that will provide health benefits to the human beings. Now come to the nutraceuticals. What are the nutraceuticals? Because the functional foods and nutraceuticals, they are correlated; there is need to understand the mechanism or the definition of the functional foods as well as the nutraceuticals. A nutraceutical can be defined as a diet supplement. It's not a food, it's a diet supplement that provides a concentrated form of bio-active compounds in a non-food matrix, to enhance health and prevent disease. There is one common example, that is carotenoids, which is found in carrot, corn and it is very essential for the maintenance of the body.

Now here we can see that the term nutraceuticals is derived from nutrition and pharmaceuticals both. But from nutrition we have taken 'nutri' part, and pharmaceutical we have taken 'ceutical' part. So by both these terms it makes nutraceutical. Now we come to the prerequisites of the function foods.

Means what properties should be there in a food to make it functional. And the very first one and the important one is that it should be a food; not a capsule, not a tablet or not in a concentrated form. And functional foods we can have; they can be part of our daily diet. And there is no need of any recommendation of the physician or a doctor to take the function foods. And another important prerequisite of this function food is that, they should contain the functional ingredients in sufficient quantities to cure the disease. Another prerequisite is that they are able to perform a particular function of prevention or curing the disease when they are taken.

Now we have discussed the function foods, we have discussed the nutraceuticals. But here it's very important to know the basic differences in function foods and nutraceuticals. If we talk about function foods they can be part of our daily diet. But nutraceuticals, they can't be part of our daily diet. They are available in the form of capsules or tablets. And we need the recommendation or we should consult a physician to take them in our daily life. We can't use them in our daily routine. For example we can say if we want to use the probiotic curd, there is no need to consult a doctor. But if we are taken any kind of nutraceuticals then we should consult and the recommended doses are also fixed about it. Secondly the basic differences are also that they are necessarily a part of our daily need as a basic foodstuff as compared to nutraceuticals. Functional foods contain nutraceuticals as one of the constituents. And these nutraceuticals when they are incorporated into basic foods, foodstuffs; they make it functional.

So this is all about the definition, their origin and what is the basic concept behind their origin and what is the basic differences in functional foods and nutraceuticals.

Thank you very much.

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Transcript

Welcome back. In the last lecture we have gone through the definitions of function foods, nutraceuticals, their origin and how we can make a food function by adding something, by subtracting or deleting the harmful substances. And in this lecture we will discuss the functional food categories. Before going to the types of functional foods, we should be aware about the different types of food. There are different types of foods and first we define that pharma or designer food. Pharma or designer foods are those foods in which some specialized food supplement is the functional ingredient, rich in disease preventing substances, and the best example of it is Vitamin D fortified milk and calcium fortified flour.

Then we come to the natural foods. Natural food as it indicates is very close to its natural form. Means nothing has been added or removed in it. And it may be minimally processed. And the example is peeled mango, as it is the best example of the minimally processed foods. Then the next category is organic foods. Organic foods we can know, we can produce these organic foods with some specific kind of production techniques in which we are not using any kind of synthetic inputs like pesticides and fertilizers. Then Whole foods. Whole foods are basically unprocessed and unrefined. And they are not containing any added ingredients like salt, sugar and oils. And the best examples of these whole foods are fruits, vegetables and unpolished food grains.

Now basically functional foods are divided into two categories; classified into two main classes. First, basic or natural functional foods. These are the foods that are naturally contain biologically active non-nutrient compound in high concentration, that provide health benefit, means these biologically active components, they are naturally present. We are not adding them from the outside source, and these compounds are mostly the phytochemicals. Phytochemicals are the bio-active compounds that we derived from the plant sources. Next is the formulated foods. And this category includes the food products especially formulated to have the higher amounts, means they naturally don't have the enough amount of that biologically active compound. They increase the amount and like phytochemicals, that are not naturally present in such a high concentration. For example we can say high fiber bread. We know that bread as such does not contain the fiber quantity enough amount.

Now the after dealing with this kind of broad classification of the functional foods on the basis of the addition or subtraction of any kind of harmful substance or enrichment or enhancement of any biologically active component. We also categorize these functional foods into various classes like a food that naturally contains sufficient amount of beneficial nutrient or no nutrient component. Example, oats; oats contain Beta Glucan. And a food in which one of the compound has been naturally enhanced. Here enhanced word we have to focus through the special growing condition; new feed composition, means we can alter, we can change the feed here and the best example in that; eggs with the increased omega 3 content achieved by altered chicken feed. Here we can alter the feed of the chicken and make it rich with the omega 3.

Then the next category is that food with the modified recipe in which we formulate or we incorporate a functional ingredient like margarine fortified with plant sterols because they don't have the plant sterols or the bio-active compounds. Next category is a food in which the nature of one or more components or their bio-availability in humans has been modified. Here the modified word is very important. By means of a specialized food processing technologies. For example we have fermented food, and in fermented food we use the specific bacteria to have the bio-active compounds or bio-active peptides which are useful to us. Next category is that a food in which if there is some deleterious or harmful component is present; either we can remove that particular component or we can replace it with a beneficial one for the good effects, and the best suited example is chewing gums sweetened with xylitol instead of sugar. Here we replace sugar with xylitol.

Then there are some examples of the conventional foods like nuts, tomatoes that contain lycopene and modified foods. Modified foods means, either we can fortify, either we can enhance, or we can enriched also and with the high folate content also and with the other supplements also. Then some medical foods are also there like PKU that is free from the phenylalanine. That is not desirable. And food for special dietary use like infant food, weight loss food, gluten free foods and lactose free food. Here I would like to make it clear that gluten free foods are very essential and lots of products now available in the market also. Actually what is gluten. Gluten is a wheat protein that may cause some type of allergy to the human being and it is named as celiac disease and this celiac disease is very harmful to the small intestine.

Similarly some kind of lactose free products are also available in the market. And these products are very suited to the people who are lactose intolerant. Lactose intolerant means they are not able to digest lactose, that is a milk sugar, because they have the lack or very less secretion of the beta galactosidase, that is an enzyme which utilize this sugar. Then types of functional foods on the basis of health claims. Means which type of health claims they are claiming on that basis we classify into two classes. Like structure and functional claims. In this it describe the effect on normal functioning of the body. But not claims that the food can treat, diagnose, prevent or cure a disease. Means it is not claiming that this particular type of food will like treat the disease, diagnose the disease or prevent the disease. And they help to maintain the cardio-vascular health and supports us immune system. Means they enhance the immune system. They are not claiming to curing of any kind of disease.

Second the disease risk-redirection claims, which imply a relationship between the dietary component and a disease or health condition. Means which component will be particularly beneficial to target or to prevent which particular kind of disease. Now there are few examples of development of the functional food products. Like for the development of functional food products, we can make a proportion between the macro-nutrients. Macro-nutrients like protein, fats and carbohydrates **(8:01)**, with the emphasis of reducing the fat content. In this we remove reduce the fat content. And which leads to the development of low fat products, that are also available in the market, like diet butter, reduced fat milk, fat free milk and low fat cheese. Few more examples of the low fat products like correlated with the health issue. If we will remove or replace fat from our food, these micro-nutrients. And if the diet constitute more calories, then it will cause, obviously it will leads to the obesity. That is the condition which further leads to the high blood pressure, diabetes, atherosclerosis and some forms of the cancer.

Then health claim regarding these are a diet with low or reduced energy intake. Here we are using two words. Low or reduced energy intake, that can contribute to a smaller risk of obesity. Then the proportion change between the micro-nutrient like protein, fat and carbohydrate with the low fat products; they are important to reduce the obesity. And carbohydrate modified with low glycemic

index. The foods which claims to be low glycemic index, they are helpful to reduce or prevent the occurrence of metabolic syndrome. New food products enriched with anti-oxidants such as carotenoids, polyphenols; they reduce
the oxidative damage on the cell membranes, functional proteins and DNA.

Here I would like to share with you the research work of my student, **Priyanka Arya** and me, we have developed a **functional bread** during her MSc program. And studied the shelf life of that bread. Actually we have used the whey in that formulation of the bread. And whey is a byproduct of the cheese and paneer industry. We have concentrated that whey and added that whey to replace the total amount of water. We have also used the banana flour, 60% and 40% of the wheat flour and concentrated whey. In that case after development of the bread, we have gone through the shelf life studies, and for the shelf life studies we have used the MAP, that is Modified Atmospheric Packaging in which we use two inert gases, like nitrogen and carbon-di-oxide. And carbon-di-oxide plays a very important role to spread the growth of the microbes. And here the result shows that these two compositions, we have used 98% nitrogen and 50% nitrogen ratio 50% CO2. And this was the best combination after modified atmospheric packaging, 50% nitrogen and 50% CO2. And when we stored our product with this combination, the product was in the good condition up to 20 days. So we have used the concentrated whey.

Whey contains the whey proteins like alpha lacto-globulin, lacto-globulin and lactoferrin. That is also come in the category of the functional foods. One more thing here. This is one of my chapter published in a book, **Functional Foods:- Sources and Health Benefits** that is published in 2017. And this chapter was based on the prebiotic, probiotic and synbiotic foods, in which I have discussed the role of probiotics, the role of prebiotics, the role of synbiotics and various kinds of synbiotic food developed by various researchers in different institutes, and what are the recent developments in synbiotic foods and the future plans for increasing the market demand of the functional foods. And also discuss the safety aspect of the functional foods. Here I would like to discuss one thing that there are some misconceptions about the functional foods or probiotics like, if we take the probiotic functional foods continuously; they will be harmful or they have some safety issues. It’s not like that. The probiotics, they enhance our immunity system. They are not harmful to us.

Then functional food trends in the United States of America. And there functional food trends are with the description of the consumer priorities. What the consumer prefers. Like speciality nutrition for the digestive health, and they describe this as the fortified foods with the more vitamins, more minerals, fish oils, omega3 fatty acids and probiotics. Then another trend is get real, very close to the natural form, very close to the nature, with the absence of artificial sweeteners, unprocessed or very less processed. Then another trend is beauty enhancing foods, which contain the natural food. Drinks, energy drinks, sports beverages or 100% juices. Another category is the protein evolution. That is more protein to maintain the healthy bones, joints and strengthen the immune system or build the muscle.

Then kids specific foods which include the nutrients and calorie level to a specific, specifically designed for the kids. Another category is the pharma food which help in the prevention of heart disease, hypertension, osteoporosis, type 2 diabetes, and also for the lowering the cholesterol. Then alternatives. These alternatives, they are free from the gluten, lactose free, meatless free substitutes like lentils and legumes. And dairy free milk. Dairy free milk means, we can use the substitutes like soya milk, rice milk, coconut milk, pea milk also. Then the sports nutrition. It is specifically for the sports nutrition supplement. Nutrition bar, energy bar or energy drink. Then weight management. Whole grains, fiber rich food. Vitamin D with more calcium. Protein, anti-nutrient and omega3 fish oil.

Here is my another chapter that I have published in a book; **Nutraceuticals Food Processing Technology** in 2017. That is, **Recent**

Trends in Functional Foods For Obesity Management. In this chapter I have discussed or I have described role of different the phytochemicals, dietary fibers, bio-active compounds. How they are important to maintain our health and to prevent the obesity. Because in obesity we have to take care about the energy balance equation regulated by the control of energy intake or energy dissipated as heat is generated.

Thank You

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Transcript

Welcome back. In this lecture we will go through the significance of functional foods in management of various diseases. Now the question arises that why there is a need of functional foods. Why we need functional foods? And there are so many factors, which we can say like the life style diseases, our way of like working, and cost of healthcare are increasing day by day which stimulate our government researchers, scientists and the food industry to have these kind of products. Functional foods also have potential to improve health, well-being and reduce the major diseases like cardio-vascular diseases, cancer and osteoporosis. So it will make a positive contribution to our health and well-being.

Significance of functional foods, they help us to improve the regular stomach and colon infections like prebiotics and probiotics. And also improve the children's life by supporting their learning capability and behavior. Overall they improve the immunity. And the functional foods they are also designed to reduce an existing health problem, such as high cholesterol and high blood pressure. Then lactose free, gluten free products are also available, that I have discussed in earlier lecture. New food products with health attributes have raised its popularity, because they are believed to offer consumer an increased ability to reduce the risk of disease.

Now health benefits of functional foods. And we categorize these health benefits in three categories like maintenance, reduction and treatment. In the maintenance side they enhance the function like gastro-intestinal health, gut microbiome like our gut microbiota, then anti-oxidants, they affect the ageing or act as a role of anti-ageing. Then improved sports performance and mental health also. In the reduction side or for a specific illness or specific deficiency disease; like calcium is very important to prevent the osteoporosis. Vitamin A is very necessary for the night blindness, then polyunsaturated fatty acids are helpful in the cardio vascular diseases. And folic acid, it is particularly required in children who have the problem in folic acid. And this is a complication that is known as Spina Bifida in which the spinal cord is not fully developed. Now the third category is the treatment or management of the illness. If we will shift the low fat products, it will solve the problem of overweight. Gluten free, it will solve the problem of celiac because it is due to the gluten content of the food product. Then low glycemic index. It will solve the problem of, it will prevent the diabetes.

Then functional foods with health benefits. Functional foods can be taken in many forms. Like some may be conventional with the bio-active compounds. And that can be identified and linked to a positive health outcomes. Some may be fortified or enhanced food, specially created to reduce the diseases risk for a certain group of people; maybe the people suffering from the cardio-vascular diseases. And the consumers can already select from a wide spectrum of food that contain functional components, like soya protein through fortification or folate fortified foods. Health benefits may result from increasing the consumption of substances already a part of an individual's diet or from adding new substances to the human's diet. Alternatively, food that do not naturally contain a substance can be fortified to provide consumers with a broader selection of food sources. Means we can also fortified if something is lacking that is beneficial to our health in the food product to have a particular health benefit, like calcium fortified orange juice.

Now functional foods with health benefits. There are various health benefits of the functional foods. Like fortified foods, juices with calcium fortification. Grains with folic acid fortification. They claim to reduce the problem of osteoporosis, hypertension and the folic acid content will reduce the risk of heart disease. And natural birth defect that is spina bifida, that I have discussed in earlier slide. Then enhanced foods are the, like in these kind of foods we enhance that particular component beneficial to us like beverages and salad dressings with anti-oxidants, phytosterols enriched flavored milk and phytosterol enriched fruit bar. And they are helpful to support overall growth. Overall health of the human being.

Especially it can be for the cholesterol reduction also. Then whole foods, like oats, they contain beta-glucan. And they are helpful to reduce the cholesterol. Then fruits and vegetables, they reduce the risk of certain cancers and heart diseases.

Now diabetes and functional foods. Diabetes is a very serious problem. And this is known as the lifestyle disease. And in this problem, this disease; the body cannot regulate the amount of sugar in the blood. And this is 2 type of diabetes. Like type 1 diabetes in which the pancreas stop producing or it will stop a very little amount of insulin, that is a hormone that regulate blood sugar level. And the second one that is the very serious one; diabetes in which the pancreas produces insulin, but the body is partially or completely not responding or not reacting to that insulin. This is a very important aspect like before launching any functional food in the market. We have to go through certain steps like first step is that phyto-nutrient analysis, means we correlate the particular phyto-nutrient which is beneficial for a particular type of disease. And after that we go for the pre-clinical screening. Pre-clinical screening means how it is beneficial to that particular disease. And for that we have to go through the clinical trials. Clinical trials are of two types. Invivo and Invitro. And then we come to the new food product development before launching into the market. Then this product will come to the market after passing through so many steps. Then product will come to the market, then we will go for the epidermiological studies, means survey kind of things how it is working, it is working properly or not. Whatever claims, are their, are performing or not.

Now to assess the functional foods. First is that we consume the food component. Then second step is that markers of the exposure to food component. Then the markers of the target function. And then markers of the intermediate end point. Here the word marker is coming again and again. Markers are the particular signals that indicate a particular level of that component. For example we can say if we are giving some diabetic person such kind of food which claim that it reduce the blood sugar. Then in that case when we take the sample of that particular patient regarding the blood sugar, then here blood sugar is a marker. Then we assess that marker; marker how it is like a by consuming of that particular food it is reducing that disease or not.

Then functional foods in cancer. Cancer is a very serious disease and trends the word cancer research from an American Institute for Cancer Research examined the relationship between the nutrition and cancer incidence in the large study on the global scale. And it was found that approximately 30% of the cancer cases worldwide could be prevented by changing the dietary habits. Means if we change our dietary habits, then we can prevent so many diseases that are known as the lifestyle diseases. One thing that here I would like to discuss, that these functional foods they are not the magic bullets. They are not targeting directly to the cancer. They are not targeting directly to the cardio-vascular diseases. They are targeting the different sides. And by targeting the different sides they have several possibilities. That includes they can alter in the characteristics of the tissues, cellular phenomena, they will reduce the infection and inflammation that will further lead to certain kinds of cancers. And they will also plays important role in the bioactive substances in the blood and tissues.

Here are some examples of functional foods. Oats contain Beta Glucan. Eggs with the increased omega 3 content that we can have by altering the chicken feed that we have discussed earlier. These both are beneficial to the heart health. Like then fermentation with a specific bacteria to have bioactive peptides. And this will help to improve the cholesterol levels. Then chewing gums sweetened with xylitol instead of sugar, and it will reduce or help to prevent the dental caries. Then golden rice and orange fleshed sweet potato with provitamin A, and we all know that provitamin A is very necessary for the vision or the healthy vision. Then functional food components available in the market. Various functional foods are available like soluble oat fiber, soy protein, phytosterol and stanol. And they claims the health benefits of coronary heart diseases. And these all like soluble oat fiber, soy protein and this phytosterols. They are Food and Drug Administration approved health claims. Then calcium which is very beneficial in the osteoporosis. It is also FDA approved; folate enriched foods. That is important for the neural tube defects that is spina bifida, I have told earlier. And it is also approved by FDA.

In this lecture we have discussed the role of different functional foods. How the bio-markers are essential. And what are the necessary steps that we have to go through before, like launching any functional food product in the market. And in our next lecture we will discuss functional food market in India and global market.

Thank You.

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Transcript

Welcome back. In the last lecture we have gone through the significance of the function foods in prevention of the diseases and what are the necessary steps to be taken before launching any new product in the market. And we have gone through the biomarkers, then importance of the biomarkers. In this lecture we will go through functional food markets in India and global market. Here I would like to mention that the world is targeting India as a market for function foods and probiotics, and lots of study material and they is available and the magazines and newspapers they are flooded with the information on the function foods, nutraceuticals and probiotics. And even the consumers they are very health conscious. And they want to these type of products; so lots of products are available in the market as probiotics and functional foods. And in this area Amul is the leading one. And they have launched the very earlier, Prolife it's a probiotic dahi and other companies are also in race like Nestle and MotherDairy.

Consumer interest in health maintenance and awareness, the role of functional food in this are the key factors generating demand. In most markets functional foods are mainly seen as food products and the factor influencing the food selection. Socio-cultural factors like availability of the local cuisine, ideology, religion, rituals, economic situation and of course the group values. Now we will discuss the consumer attitudes towards the functional foods. Means how easily they are adapting these function foods in their daily life. Here we can see that how frequently the consumer purchase the foods that promote the specific health benefits. Here in this figure we can see that approximate 10% of the people, they never bought whole grains high fiber products. Similarly if we see here, like more than 15% of the consumers, they don't have the trend of buying the yogurt with acidophilus culture or probiotics. In case of iodine enhanced cooking salt, the figure is high that is more than 20%. Cholestrol reducing oils and margarines, the figure is quite high that is approximate 20%. And the consumer who never bought these kind of products like fermented drinks containing good bacteria are approximate 25%. Fruit juices with added supplements are the consumers. They approximately 25% of the consumers they never bought these kind of products.

Then bread with added calcium or vitamins. Here you can see that more than 25% of the consumers didn't have the practice to buy such kind of products. Then milk with added supplements like vitamins and other supplements. The percentage is quite high, that is approximate 30% of the consumers are not in this kind of practice. If we will see the cereals with added folate, approximately we can see 25% or more than 25% of the consumers they are not in practice to adopt or to buy the cereals with added folate. Now this is regarding soy milk, and here we can see that more than 35% of the consumers, they never purchase these health promoting foods that is soy milk. One thing that I want to make clear here that soy milk is a very good substitute forthe people who are suffering from the lactose intolerance problem. And if we compare about the cost, soy milk is comparatively very nutritive, comparatively cheap as compared to the normal milk.

So the consumers have a misconception about the cost like if we say probiotic food; function foods, the consumers immediately think about the cost, means it will be a costly product. It is not like that the function products or the probiotic products may be costly, may not be costly, it depends upon the nature of the food products. Like if you will compare the probiotic dahi with the normal dahi the cost is comparatively like not so much difference is there. Functional foods on the global market; which foods are available globally. Like probiotics, prebiotics and functional drinks. Here probiotics, I would like to discuss that probiotics are the live microbes, when taken they confer the health benefits. And in the simple language we can say that these are the good microbes. They enhance our gut microbiota, and examples of these are the 2 species that are very frequently used; lactobacillus and bifido bacteria.

And after them there is a correlated term that is prebiotics. Prebiotics are in very simple language, we can say the feed or the food of the probiotics. Means if we add any kind of prebiotics like inulin that is derived from the chicory roots or fructo-oligosaccharides, that is FOS. If we add this during the preparation of the prebiotic food, they will enhance or they will stimulate the growth of the microbes, because they are live. Then functional drinks, they are helpful in weight control. Nutrition beverages, energy drinks, sports beverages, and ready to drink tea and coffee. Functional cereals, oat based products, barley based products. Because oat contains beta-glucans, that is very useful. Functional meat, meat products with added functional ingredients, vegetable proteins, dietary fibers, herbs and spices. And dietary fibers are also important to mention here that they entrap the harmful toxins that are produced in the body and prevent from many kind of problems.

Then meat products which are modified during processing. Like during processing or production. During production we can add some kind of microbes which release the bioactive peptides during the fermentation or during the curing also. Reformulated meat products. Reformulated means we can reformulate by addition or by subtraction like fat reduction, cholesterol reduction, reduction of sodium and nitrite levels. Improvement of fatty acid composition also. Then enriched eggs. Eggs enriched through the supplementation of the animal diets with the functional ingredients like long chains polyunsaturated fatty acids, vitamin E, selenium, conjugated linoleic acid and lutein.

Now the key market drivers that is regarding the functional foods and nutraceutical products in Europe on the basis of the health claims. Means which kind of health claims they write on that product. And the most desirable health claims in Germany are like boost immunity system, promotes healthy bones, promotes healthy teeth, gives energy, promote healthy gut. Means these are the desirable claims, that the product sold in Germany. They have the these kind of desirable claims. Similarly in U.K. the claims are different, like gives energy, promotes health bone, promote healthy teeth, reduce risk of cancer and lowers the cholesterol. In France the desirable claims are like; gives energy, lowers cholesterol, increases disease resistance, boost immunity system and prevent constipation.

International market for function foods. What is the scenario of the functional foods in the international market. Internationally functional food market has experienced remarkable growth. And in India also they are contributed to the economic growth of the country and added revenue to the country. This is one of the most dynamic sector in the food industry, because they have the latest concept, the latest product. The recent products in the in the term of product development and international expansion. Means lots of new products are coming day by day in the market. But the functional foods and probiotics, they are booming in the market. While large food companies they are aggressively harness larger market share, small and medium enterprises, that is SME's; they are also successfully identifying untapped opportunity and building up the market niche.

So here we have discussed, we have moved through the different aspects, the consumer behavior to other function foods. What is the market scenario? What are the different kind of health claims in the different countries, in this lecture.

Thank You.

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Transcript

Welcome. This lecture is about the regulatory systems for health foods. And the regulatory system or regulatory aspects is very important to discuss here, like there are so many regulatory bodies who regulate about the labelling part, the health claim part and the right doses, because the consumers they are health conscious. They should be aware about the right doses of the phytochemicals, the right doses of the nutrients, which could provide the health benefits to them. So the regulatory bodies they are working in this area regarding the nutritional claims and the health claims, and the advantages and the labelling should be proper, that is easily understandable by the consumers. Safety issues of functional foods are becoming very popular day by day throughout the world. And the safety issues are also triggering brains of the researchers as well as the consumers, because today's consumer is very health conscious. They want to know the right doses, the harmful aspects of the these kinds of foods. It is very important to determine the right doses. I have earlier discussed this point that right doses should be determined. How much it should be taken regularly or daily of the bio-active compounds or phytochemicals. Phytochemicals are the plant waste bio-active compounds that we derive from the plants. That to be incorporated into the food system to be administered as nutraceuticals.

Then safety scale of functional foods and nutraceuticals. There is a very fine line between the safety scale of the functional foods and nutraceuticals. Functional foods they are comparatively safe as to the nutraceuticals. And in case of functional foods there is no restrictive specific limit of consumption as compared to the nutraceuticals. Provide slow results, but the nutraceuticals they provide fast results. Functional foods can be consumed as a basic foodstuff along with the main meals. Earlier also we have discussed that the functional foods have no like no particular amount or specified amount. For example probiotic dahi or probiotic yogurt, we can have in our daily diet, but without any prescription. But in case of nutraceuticals we have to consult how much doses are necessary, the right doses and the duration of the doses also, because they are in the concentrated form. The phytochemical and the bioactive compounds they are in the concentrated form in the form of capsules etc.

Then functional foods they are very diverse. And minimum standards are there for their preparation. But in case of nutraceuticals, the high standards for their manufacturing are required. Similarly with the prescription part also. Then minimum education is required for the regulation or the sale. But in case of nutraceuticals, it is strictly; like an educated person or the experienced person in that particular field. Only that person can suggest or derive the regulatory aspect for the sale. In case of functional foods we don't require any medical advice before consumption, that we have discussed. But in case of nutraceuticals a medical advice is needed because they are in the concentrated form.

Regulations related to the functional foods. The regulatory bodies like Food & Drugs Administration(FDA) under the authority of two laws. First is Federal Food, Drugs and Cosmetics Act, 1938 (FDAC) provides for the regulation of all foods and food additives. And the second one, The Dietary Supplements Health and Education Act of 1994, and it will cover the dietary supplements and the ingredients of dietary supplements. Now another regulatory Codex Alimentarius. And this regulatory body has defined two types of nutrition claims. That is nutrient content, nutrient content means that is a claim that describe the level of a nutrient contained in the food. Means that particular food contains how much of that nutrient. Then second is nutrient comparative is a claim that compares the nutrient level or the energy value of two or more foods. Nutritional content claim. It means any representation, which states, suggests or implies that food has particular nutritional properties including but not limited to the energy value only, and to the content of the protein, fat, carbohydrate, as well as the content of vitamins and minerals.

Nutritionals claims. First we should know that what are the nutritional claims. Like whenever we have visited or you also have visited some malls or some markets for the purchase of different kind of things. We are searching for the new products, and first of all whenever we pick any kind of new product. What we look for. We go for the ingredients, what it contains. And the health claims. Means, we go for the if somebody has the diabetes problem he or she will find out the claim that it is sugar free, low sodium, fat free, lactose free and that is gluten free. So nutritional claim means any claim which states, suggests or implies that food has a particular beneficial nutritional property due to energy value, and also provide at a reduced or increased rate. Does not provide the nutrients another substances.

So different nutrition claims like low energy, energy reduced, energy free, very low sodium, sodium free or salt free. That its source of fiber. That which kind of nutrient it contains. It also comes under nutrition claim. Then increased, increased means; any particular component or any particular nutrient has been increased or decreased or removed from that food. Low fat then fat free, low saturated fat, saturated fat free, low sugars, sugar free, then with no added sugar, low sodium, source of fiber, high fiber, source of protein and high protein. These are all the nutritional claims that the regulatory bodies say that it should be mentioned on the label, the nutritional claim. Like then we will come to the source from where it is derived; plant based, animal based or any other source, name of the mineral if it has mineral. Which mineral is incorporated like calcium, iron or some other mineral. Then high means the name of that particular vitamin or mineral that is added; increased which nutrient is increased. It should be mentioned. Then reduced, name of the nutrient, which nutrient has been reduced. Then lite term is also there. Then naturally or natural that is related to the nutrition claim, like naturally high in fiber, we always search, the people who are health conscious always search for the product that are high in fiber.

Rich sources of omega 3 fatty acids. Because they are very essential for the brain development. Then high omega 3 fatty acid. Then high mono-unsaturated fat, high poly-unsaturated fat, high omega 3 fatty acids. And no added sodium or sugar. And then reduce. Reduce term means any kind of nutrient or any kind of component that has been reduced, increased or enhanced. It is mandatory to claim that nutrition, nutritional part. Similarly it is about the health claims. Health claims means any claims that suggests or states the relationship that exists between a food category and a food or one of its constituents and health. Three things are there. The food category, the food or its constituent and the health. It's a trio.

Then health claims are mentioned differently. First we can say that the impact of a particular nutrient to the growth and development of the organism. What is the impact of that particular nutrient that we are adding or removing in the food; functional food. Secondly the impact on the physiological function, behavior, weight control or reduce the feeling of hunger or increase the satiety or reduce the energy availability. Third aspect is the impact of health and development of children. Then fourth aspect is the risk reduction of the disease. Means that health claims will, claims like it will reduce some particular kind of disease or not.

Functional food regulations in India. What are the regulatory bodies in India? There is a big concern for the consumer as well as the government and regulatory bodies. And FSSAI is the recent one that works on these regulatory aspects of the functional foods and other foods also. In FSSAI chapter 4, article 22 of the Act, it addresses about the nutraceutical, functional foods, dietary supplements and the need to regulate these products that any one can manufacture, sell, distribute or import these food products. It covers all these parameters. These food products covered under it include novel foods, genetically modified articles of the food, irradiated food, and in this kind of food we use irradiation. Then organic food and food for the special dietary uses. Functional foods, nutraceuticals and health supplements. There is article 23 & 24 addresses about the packaging and labelling part of the food and restriction on the advertisement regarding the food. Here packaging and labelling part are very important. The packaging material we are using, because these are the food that is related to our health. And the labelling aspect is also important. Because the consumer, he is not aware about the claims that written on that particular food product, so the labelling part is very important. It's very essential to label that particular claim on the food product.

Then there are international regulations also. Different countries have different regulatory bodies. In Japan it's FOSHU and it approves the statement made on the food labels concerning the physiological effect of the food on the human body. Then United States of America. Food & Drugs Administration, that is FDA regulated the nutraceutical industry, since long, since 1947. And health claims may be authorized for a food based on an extensive review and the scientific literature. And then Canada. Health Canada regulates the health food industry. And the Canadian Food Inspection Agency enforces these regulations. So most of the nutraceuticals in Canada fall under Natural Health Products

Regulations of the Food and Drugs Act, which came into effect since January 2004. In New Zealand, Food Standard Australia and New Zealand. And in China, Food hygiene law of the People's Republic of China. And the administrative licensing law of People's Republic of China. And this law is effective there from July, 2005.

So these all about the regulatory body. In this week we have discussed the different aspects, the different definitions of functional foods, their significance, the role of regulatory bodies, the need of regulatory bodies and the different sources. Their mechanism affection. The importance of the functional foods to prevent the different diseases and to provide health benefits. Next lecture we will discuss, functional food categories and sources.

Thank You.

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Transcript

Hello everyone, welcome to week 2. I hope you have gone through the contents of 1st week, and you have find something important in that content regarding the basic concepts, health benefits and other contents. So let us start with the 2nd week content. And 2nd week contents have lots of interesting topics. And the topics we will cover in 2nd week are,

- Plants and Animal Based Source Functional Foods
- Benefits of Functional Components
- Bio-active Compounds and their Role in Health Promotion
- Cereals and Vegetables as Functional Foods, and
- The Status of the Functional Foods and Availability in the market

When diet is wrong, medicine is of no use. When diet is right, medicine is of no need. This is very rightly said. By changing our eating habits, and by knowing the beneficial aspects of this bio-active compounds, we can change our lifestyle, we can enhance our health and can prevent many serious diseases. Like in every time we are talking about the functional compounds; the functional compounds. What the functional compounds are? How we will define that? Functional compounds are the biologically active compounds that are found in food, and they have the capability to alter or to change the different processes, the metabolic pathways in the body. And that's why they are responsible for the alteration or for the prevention of the diseases and promote our health.

Now let's start with our first lecture that is 'Plants and Animal source Based Function Foods'. Now the functional foods that we derive from the plant sources are

Primary metabolites, that includes the

- plant proteins
- beta-glucans, and
- omega-3 fatty acids.

Secondary metabolites, these include the

- phytoestrogens
- antioxidants
- tocopherols
- steroids, and
- gamma-linolenic acid that is GLA and
- phase 2 enzyme inducers, and
- brassica vegetables that include cabbage, cauliflower.

They act as a functional foods. And their mechanism of action is like, they are able to glycosylating the insoluble toxins to produce soluble compounds that are excreted. Now animal derived functional foods. We call the animal derived functional foods as zoo chemicals. And the functional compounds that we derive from the plant sources, that we call the phytochemicals. So the zoo chemicals they include the

- omega 3 and omega 6 fatty acids,
- GLA that is conjugated linolenic acid,
- small peptides,
- whey and casein, and
- glucosamine.

Whey and casein are the milk proteins. Omega 3 fatty acids include alpha-linolenic, DHA, EPA fatty acids and others. Now microbial sources are also important to mention here as the source of the bioactive compounds or function compounds. There are three sources of these function compounds. First the probiotics, and probiotics are the live micro-organisms, when administered they confer the health benefits. And two species are very important to mention here; that are generally regarded as safe, as a grass. That is lacto-bacillus and bifida bacteria species.

Secondly the prebiotics. Prebiotics are the known digestive substances, that enhance or that stimulate the growth of the probiotics. Means if we add some prebiotic substance to the probiotics, they will stimulate the growth, they will enhance the growth. And third aspect is synbiotics. Synbiotics means combo of probiotics and prebiotics. And the word synbiotic is, I will lots of meaning here. Like the relationship should be synergistic type, means mutual understanding should be there, means one will enhance the growth of the other. Some other kind of functional food sources are also there. Algae, it will provide the omega 3 fatty acids; which enhance the immunity, also modulate the inflammation. And protect the neuro-degenerative diseases. Mushrooms are also good source of functional compounds. And they are, by providing the beneficial effect like anti-viral, anti-bacterial, anti-inflammatory properties.

Now genetic engineering and function foods. Have you heard, I think about the genetic engineering concept also. And what is the importance of genetic engineering in functional foods. Genetic engineering, we go through the cross production, means the desirable gene. We take a desirable gene, that we want for a particular benefit. And we take that gene and insert in that particular plant for the beneficial effect. But there are some resistance regarding the acceptance of the genetically or GM foods or GM crops. A few example in case of functional foods are like. We can enhance the components by improving the fatty acid profiles in oilseed crop, and we can also modify the protein quality of the potatoes, and anti-oxidant content in different crops by adopting the genetic engineering.

There are few examples to make you clear about the genetic engineering aspect. Like rice. Rice did not have the pro-vitamin A; that is very good for the healthy vision. What we have done during genetic engineering, like we use the transgene, that is phytoene synthase, from the Daffodil. And phytoene desaturase from the Erwinia. We take the particular trans-gene and we use this gene in the rice crop to enhance this particular substance, pro-vitamin A. That is very beneficial, I have told you in the healthy vision. Similarly we will talk about tomato. In case of tomato, we have a trans-gene; we take the transgene, that is chalchone isomerase from petunia; and we insert this gene for the flavonoid contents. And flavonoids are you know that, are the potent anti-oxidants. And the third example is rice. In rice if we want to enhance iron content, then ferretin that is derived from phaseolus; that is a trans-gene, act as a transgene. We will insert this trans-gene in the crop production, during rice production. And it will enhance the iron content and use it as a iron supplement also.

So this is all about this lecture. Like the different aspects, the different sources. Plant based sources, animal based sources. Even the microbes, they are also important as a bio-active compounds. And the genetic engineering and it's role in the functional foods. This is all about this lecture.

Thank You.

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Transcript

Hello everyone. We have seen that there are many sources of bio-active compounds, like animal based sources, plant based sources. And even the microbial sources are also there. So let us start our next lecture, that is based on benefits of functional compounds in food and health promotion. Functional ingredients of animal based products, that is first very important class is fatty acids. And fatty acids include CLA, that is conjugated linoleic acid, and the sources are milk, cheese and meat products. And they are very beneficial and improve the body composition, decrease the risk of certain kind of cancers. Salmon, tuna, mackerel and fish oils; they are very rich sources of long chain omega-3 fatty acids, that is DHA and EPA. And they are claimed to reduce the cardio-vascular diseases and improve the mental and visual functions.

Second important class is plant based foods that is polyphenols. Polyphenols include

- anthocyanidine
- catechins
- flavonones
- flavones, and
- tannins

Fruits are the sources of anthocyanidine and catechins are found in tea, babul pods, mustard seed and rapeseed, and flavonones, they are also found in citrus fruits, vegetables and soya beans. And all these polyphenols, they neutralize the free radicals. You know free radicals are harmful to us and reduce the risk of the cancer. Then tannins. The source of tannins are cocoa, chocolate and cranberries. And they reduce the cardio-vascular diseases and improve the urinary tract health. Another source of these bioactive compounds are the saponins, and they are derived from the soya beans and chickpea. And the health benefits, they cause and lowers the cholesterol and anti-cancer.

Then probiotics, prebiotics and synbiotics. These are of the microbial origin. Probiotics, lacto-bacillus that is present in the fermented products; yogurt and curd. And prebiotics, that is FOS(fructo-oligosaccharides) or we can say inulin also. And the sources are whole grains, onions, combination of probiotics and prebiotics; that is synbiotics. And they improve the gastro-intestinal health and improve quality of intestine microflora. Then next class is phytoestrogens. It includes isoflavones; that is diadzein and zenistein and the sources are soya bean and soya waste products. And they claim to reduce the menopausal symptoms like hotflashes and also increase the bone health. Lignans are also very important and the sources are flax, rye and vegetables. And they reduce cancer, heart diseases, and lowering the low density lipoproteins and total cholesterol.

Next is carotenoids that are very important. And carotenoids it is alpha and beta carotenoids. They are found in papaya, carrots, vegetables, fruits and especially mango. And they neutralize the free radicals. Then luteins, that is found in green vegetables and maintain the health and healthy vision. Then eggs, citrus fruits and corn. They are the good source of zeoxanthine. And tomatoes are the good source of lycopene. And they reduce the risk of prostrate cancer. Then come to the dietary fibers. It's important to know that we have heard that dietary fibers are a very important, very important and should be included in our diet. Do you know how they work? How they are beneficial to us? Let's see. Dietary fibers which could be insoluble or soluble are starchy polysaccharides and structural components of cell walls of cereals and micro-organisms. They are the indigestible part of the plant foods and compounds of long, straight and branched chains of carbohydrate molecules held together by the bones that cannot be hydrolysed by the human digestive enzymes, means the human digestive system enzymes, they can't act on the dietary fibers.

The long fibrous structure of dietary fibers allow them to trap the harmful toxins and carcinogens. Carcinogens means which are responsible for cause of the cancer. So these long fibrous structures they entrap the harmful toxins as well as the carcinogens in the digestive tract. That's why the inclusion of dietary fibers in our diet is very important and this is how they work for the beneficial purpose. Dietary fibers are two types, like insoluble dietary fibers, that is found in wheat bran, food grain and husk mainly. And soluble dietary fibers, that is beta-glucan found in oats and barley. And oats, like these days are very popular because of the beta-glucan and this is a soluble dietary fiber; that is very beneficial, I have told you earlier. And they will helpful, the dietary fibers, they are helpful to reduce the cardio-vascular diseases. And the dietary fibers are also found in the whole grain, that is the cereal grain.

Now once again I would like to discuss with you like the mechanism of action of the bio-active compounds. The food have the different kind of bio-active compounds. It may be present in the foods. It may be present in the vegetables. Then how the bio-active compounds work. The important thing is to know the mechanism, the science behind it. So let's see how they work. These bio-active compounds, they work as the substrate for the bio-chemical reaction. They provide the substrate for the bio-chemical reaction. Secondly, they are the cofactors for the enzymatic reactions. And another way of working is that, they inhibit the enzymatic reactions somewhere, and they act as a absorbents or that bind or eliminate the undesirable constituents in the intestines. That is not desirable, that is harmful. They bind that constituent.

Another way of working is that they act as a scavengers of the reactive or toxic chemicals. And they also play an important role in the growth factors or substrate for the probiotics, because I have told you that some substrate will be there. It will enhance the growth of the probiotics. So this is all about the mechanism of the bio-active compounds. How the dietary fibers are useful in our diet? Why we should include these dietary fiber in the diet? And why the beta-glucan is important. And various other sources.

Thank You.

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Transcript

Hello welcome. I hope you are enjoying the course with me. And I assure you that after completion of this course, you will start look the food beyond the basic nutrition only. Let's start with the new lecture, that is role of functional compounds and bio-active compounds and their role in health promotion. First important aspect is the dietary fibers that soluble and insoluble form. The examples are

- beta-glucan
- gums
- pectin, and
- mucilage

They are the water soluble dietary fibers.

- lignin
- cellulose and
- hemi-cellulose

they are the water insoluble. They perform the different functions. How they perform the different functions? How they act? And how they cause benefit to the human beings. Like they entrap the harmful toxins and carcinogens that cause cancer in the digestive tract. They are also known for their good water retention capacity, gelling ability and hydro-colloidal forming properties, which have influenced their use as a substitute of fat. They can be used as a fat substitute also. Now in insoluble dietary fibers, they cannot be dissolved in water as the name suggests. And they are effective in adding faecal bulk, and increasing the rate of passage of food through the intestinal tract.

Now what are anti-oxidants and how they works. What is their role? What role they plays for the health benefit? Anti-oxidants, how we will define. Anti-oxidants are group of compounds, which neutralize free radicals. Free radicals are harmful to us and reactive oxygen species, that is ROS in the cell. And various functions that are performed by anti-oxidants. They include the regulation of redox potential within the cell. And reduction of potential initiators of the cell death and carcinogens. They increase the cell permeability and cause the cell death and remove the carcinogens or delay the carcinogenesis. Now what different diseases or damages are caused by the reactive oxygen species, that is ROS; that we have told earlier. Like reactive oxygen species, they are very harmful and responsible for causing many serious diseases like diabetes, infections, inflammation. And inflammation leads to form, take the other form of cancers and the other problem like arthritis and inflammation and ischemia, Parkinson's disease, radioactive damage, shock and increase; fasten the ageing process and arthritis also. They cause arthritis also. So they are very harmful and damage to the human beings.

Now come to the free radicals. The term this free radicals is also coming again and again. Free radicals are produced, free radicals are generated. Then what are these free radicals. And how they are important or how they are like damaging the body cells. Let's see what are the free radicals. Free radicals can be formed in lipids also; protein and carbohydrates. Actually free radical is a carbon or oxygen atom that has an unpaired electron and is highly charged. And this unpaired electron is very unstable. Here we can see that these are the free radicals. And they are found in the proteins, vitamins and the carbohydrates. And these free radicals, they act on the body cells and cause damage to the body cells. Then which kind of the damage they cause. Like the damage which is related to skin, it may cause wrinkles, and arthritis that is related to the joints. In case of heart diseases, brain related to the it cause the Alzheimers disease. And they destroy the DNA which leads to the cancer.

Now probiotics and prebiotics. Probiotics are generally believed to be protect us, like working in 2 different ways. First role that they play in our digestive tract. Means they enhance the good bacteria or they enhance our micro-biota of the digestive system, that we are talking about the good microbes. And these good microbes, they inhibit the growth or entry of the pathogens that cause diseases. The second major benefit of the probiotic is that, they boost our immune system. Omega-3 fatty acids and their health benefits. Lots of researches has been done related to it and the omega-3 oils, they have a very proven health benefit. Long chain omega-3 polyunsaturated fatty acids. They are of great importance because of the prevention as well as the treatment of the coronary heart diseases, hypertension, diabetes, arthritis, and another kind of some inflammation, auto-immune diseases; like ankylosing spondylitis, mental health, cancers. And they are also important for maintenance and development of normal growth, especially for the brain and retina. Omega-3 fatty acids are very important for the development of the brain and retina. And lots of research has been proven to this effect.

Then the different, this omega-3 fatty acids, or they are available in different forms. Like triacylglycerons or tryglyceron concentrates. Then ethyl esters or its concentrates. EPA and DHA, phospholipids, calcium and magnesium salts, chromium in DHA complex form, phytosterol, DHA ester, and epigallocatechin, that is EGCG-DHA ester. Then the phenolic and poly-phenolic compounds. How they works to prevent the different problems or different diseases. Like they remove the ROS, that is reactive oxygen species; earlier we have discussed, of the cell and they also works as a anti-oxidant. They affect the cell differentiation. They increase the activity of carcinogen detoxifying enzymes. These enzymes, they detoxify the carcinogens, which cause cancer. They also block the formation of N-nitrosamines, that is harmful. They also alter the estrogen metabolism, also increase the apoptosis; that is the death of the cancer cells because of the cell permeability, and thus decrease the cell proliferation.

And these poly-phenolic compounds they also affect DNA methylation and maintain the DNA repair. And thus they are also preserving or important in preserving the integrity of the intra-cellular matrices. So this is all about the role of bio-active compounds. What are the free radicals? How these free radicals generated. And what the roles of the anti-oxidants in this free radicals. Obviously the anti-oxidants they neutralize the free radicals that are harmful.

Thank You.

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Transcript

Hello I hope you are doing well. And as you know that fruits and vegetables and grains, like whole grains; we call cereals, they are also important as a functional foods. Let's see how. Cereals and vegetables based functional foods. Phenolics, polyphenolics and antioxidants. And the fruits you know contain the phenolics and polyphenolics. And mainly these phenolics and polyphenolics, they are found in leaves, skins of the fruits and seed part. Actually leaves are a very big source of this polyphenolics as an antioxidant. Then cereals and legumes, the bran portion especially. That's why we call bran part as nutrition. And the bran course is very enriched in this polyphenolic content. And as compared to the endosperm. Means endosperms as comparatively have less phenols as compared to the bran portion. Then they are very important for the as an antioxidant part, due to their role and efficacy in scavenging free radicals, means they remove the, scavenge the free radicals part that are harmful. They also chelate the pro-oxidant metal ions and acting as a reducing agent.

This phytochemicals. Phytochemicals, they are also present in vegetables and cereals also. And they don't have only one particular benefit. They have lots of health benefits. The phytochemicals act as a substrate. They act as a cofactor for the enzymatic reactions. They act as inhibitors for certain kind of enzymatic reactions, where required. And they also worked as a absorbents, and eliminate the undesirable constituents in the intestine. And they scavenge the reactive or toxic chemicals that are harmful. They enhance the absorption of the essential nutrients required for the body. And they also plays very important role for the as the growth factor for the beneficial bacteria. That is we can say the probiotics. Then they also act as a substrate in the fermentation for the beneficial bacteria. Substrate means they act as a food; means the beneficial bacteria they can, then use for the energy and other purposes, and mainly for the growth. And selective inhibitors, they remove the deleterious components from the intestine.

Now we will talk about the sources and health benefits of the phytochemicals. From where we derive these phytochemicals and what kind of health benefits they provide. Grains, basically they are a rich source of tocotrienols and tocopherols; specially they suppress the growth of diverse tumors cell lines, and cause the cell death. Then fruits and vegetables they contain carotenoids. And carotenoids we have discussed, you are also aware that these are the potent antioxidants. And protects against the uterine and prostrate cancer, colorectal, lung and digestive tract cancers. And protection to the other antioxidants also. Now citrus fruits they are rich in limonoids. And these limonoids, they inhibit the phase 2 enzymes. And also the phase 2 detoxification of the enzymes in liver and provide protection to the lung tissues. That's why they are beneficial. Then various plants they contain phytosterols also. They are good source of phytosterols also. Then phytosterols, they are beneficial. And also exhibit the anti-inflammatory, antineoplastic and anti-pyretic immune modulating activity. That decreases the cholesterol level.

Then various plants and whole grains, they contain the phenolic constituents. And these phenolic constituents they act as a antioxidants and reduce or lowers the risk of coronary heart diseases, diabetes, hypertension and others. Similarly the fruits and flowers, they are the rich source of anthocyanidines. And anthocyanidines are good antioxidant properties, have good antioxidant properties and anti-mutagenic properties. Cruciferous vegetables like cabbage, cauliflower, broccoli. They belongs to the cruciferous family and they contain glucosinolates. And they act as a liver detoxification of the enzymes. They inhibit neo-plastic effect of various cancer causing factors. Now vegetables and fruits, they also have fiber. And now you have understand the role of fibers in our diet. They also protect against the colorectal diseases.

Now here you can see that the cereals with their antioxidant activities. Here we have taken the materials like different cereals. They have total phenolic content that is in milligram, ferulic acid equivalent per gram, that is the unit how to measure it. Like antioxidant activities in micro mol (μ mol). Like here you can see that; wheat, varieties of wheat. But this is bran part and this is flour. Wheat flour and wheat bran. Here you can see that bran part is comparatively very high in total phenolics that is 66.91 as compared to wheat flour, that contains 24.1. The difference is too much. If we see the antioxidant activity; wheat bran it shows 55.80, and comparatively wheat flour shows a lower one that is 27.10.

From this we can conclude or we can see that the bran part is rich in which part, that is total phenolics as well as the antioxidant activity. So bran part is important here. This is the comparison in the wheat part. And that is wheat bran and wheat flour. Now come to the barley. Barley is also a cereal. This is the outermost layer and this is the innermost layer. And the comparison is that the total phenolics are 6.26, and in innermost layer the total phenolics are only 0.51, that is very less comparatively. But in case of antioxidant activity also, the outermost layer, it contains the 59.70 and innermost layer it contains 0.45. You can see the difference here. It is a great like difference. Together we can conclude that the outer layer is very rich in total phenolics as well as the antioxidant.

Let's come to the millet. And kodo is a type of millet. Millets are grown in our country also, like pearl millet, kodo millet, sava. This kodo is one type of millet. And here we have taken whole grain soluble, this part. Then same variety, of millet that is kodo; dehulled grain soluble. And third is kodo with whole grain bound. Three aspects are there. Let's see, in whole grain soluble the total phenolics are 32.40. That is higher one. Then dehulled one 6.86. Dehulled means we have removed the hull from that part or dehusking or dehulling has been done. We have removed the hull from here, so that we are calling it dehulled. It contains 6.86. And kodo whole grain bound not soluble one. Whole grain it contains 81.60. Over here you can note the difference. The whole part comparatively has the highest one, total phenolics, that is 81.60. Now come towards the antioxidant activity. In case of kodo whole grain soluble. Soluble part is the richest one, in all three in case of antioxidant activity. That is dehulled, it has comparatively 41.00. And kodo whole grain bound, that is total antioxidant activity, 79.30. So we can compare the whole grain soluble here as a highest antioxidant activity. So this is all about the antioxidant activity of the wholegrain.

So in this lecture we have discussed that various sources of the antioxidants derived from the vegetables and the whole grains. And I hope you are enjoying the course.

Thank You.

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Transcript

Hello, welcome back. By now we have studied different aspects of bio-active compounds. Cereals and vegetables as a functional compounds. Their role and their significance, different health benefits and many other things. Now in this lecture we will discuss about the different market availability of the nutraceuticals and functional foods. Have you visited the market to buy some kind of functional foods or nutraceuticals. Do you know which type of products are available in the market? Now let's have a look. Some examples of the functional foods. Like oats. Everybody is aware of the nutritional significance of oats, which contain beta-glucan and omega-3 enriched eggs. Both are good for the heart health. Then fermentation with a specific kind of bacteria, that is the grass bacteria. We can say it is recommended and say, that is lacto-bacillus species to have the bio-active peptides. And they improve the cholesterol levels.

One other example is chewing gums. That they will be replaced sugar with the xylitol. And it will prevent the dental caries. Then golden rice and orange fleshed sweet potato with the pro-vitamin A, that is good for the vision. Nutraceuticals that are available in the market. This is the brand name. And this is the particular component present, which is responsible for targeting the health benefit or targeting to prevent the disease. And what is the function of this component. The brand name is betatene, and it contains carotenoids. Carotenoids, they are antioxidant, you are aware about that. And it will improve the immune function. Similarly one more product is there, Xangold. These are the brand names. I am telling the brand names. When you are searching in the market, you should be aware about the brand names.

And what is the particular component? If we have knowledge of the particular component. What it contains actually, only then we can come to its function. Like it contains Lutein esters. And it is good for eye health. Lipoec, it contains alpha-lipoec acid. And it is a very potent antioxidant. Then Generol, it contains phytosterol. And it helps in prevention of coronary heart diseases. It will reduce the coronary heart diseases. Then the another brand, that is a popular one. The Premium probiotics. And the component are probiotics. Till now you have come about, know about the probiotics. What they are? How they are useful to us? And they improve our immune system. And intestinal disorder because they make the nutrients unavailable to the pathogens. Pathogens are the disease causing micro-organisms. And probiotics are the live microbes that confer the health benefits. In the simple language we are calling them the good bacteria.

Then another important product that is available in the market is Soylife. And it has the soya bean phytoestrogen. And this is good for the bone health. So we should be aware about the brand and the particular component present in that. And that component is targeting which function. Are the nutraceuticals that are available in the market. Similarly lots of nutraceuticals are available in the market with the functional component. This is a functional component. And this is a function of that particular component. The brand name is Z-trim and the component is wheat. It is a zero calorie fat replacer. Then Linumlife, it contains lignin extract from flax and it is beneficial in prostrate health. Fenulife, that is it contains the fenugreek, in normal language we call it methi and it controls the blood sugar. So these products are, these products in the form of nutraceuticals, they are available in the market. Next is Teamax. It is a green tea extract. And it is a potent antioxidant, which helpful in the scavenging of the free radical.

Then Marinol. It contains DHA and EPA. Omega-3 fatty acids, and it is good for the heart health protection. Then Clarinol. The active compound here is CLA. It is conjugated linoleic acid. And it is useful to maintain the weight loss of the ingredients. Then Cholestaid. The active component or the bio-active component is Saponin. Saponin you are aware that it can reduce the cholesterol. So now after discussing all these things, you are aware of the functionals and nutraceuticals that are available in the market. Whenever you visit the market or any place for buying of these products. You should be aware with the health benefits and its active components. That are the main importance. Now we have discussed the market availability also, the benefits also. How they are useful? How they can prevent the disease? How they promote the health?

Now there are some constraints also regarding the consumers. The consumers like they have some kind of hesitation in buying of these products. And with this hesitation there are other factors also, which plays; like, act as a constraint in buying these products. What are these constraints. Let's have a look. First is the low income of the huge population. If we talk about our country. If we'll talk about India that the income group matters a lot. The low income people, they have the conception that these products are very costly and they are not able to afford them. And huge population suffer with this type of misconception. Then second factor is existence of the unscrupulous manufacturers. Means the manufacturers, they are not valid. And another constraint or another factor is the testing infrastructure. Means whatever products are manufactured by any company. They are lacking in the. There are no infrastructure for testing them. And also to validate the manufacturer's claims. Means the manufacturer if that company or manufacturer is claiming that sugar free. It is a claim. So there should be some authority that can validate that really it is free from sugar or it is a sugar replacer.

Then the important other aspect. It is the physical infrastructure that is also a major constraint in the manufacturing of the functional foods. Then another regulatory framework for functional foods. Now, these days FSSAI is a very regulatory body, who is looking after these functional foods and nutraceuticals. And I think all kind of products FSSAI is looking for. And another more important factor is, resistance to the genetically modified foods. Earlier also I have discussed that we have a mindset that genetically modified foods are not good. And we till date we have not accepted them. So people's are then, they are like very conscious and very resistance they have for the purchasing or adoption of genetically modified foods.

So this is all about the market availability of the products. And the different factors which affect the buying capability or buying capacity of the consumers, and the health benefits. Which brands are available. And while purchasing any kind of nutraceuticals or function food, how we can go through the health claim, and other aspects.

Thank You.

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Transcript

Hello welcome to week 3. I hope you have gone through the contents of 2nd week and 1st week, and you have gathered a very updated knowledge and information about the Functional Foods and Nutraceuticals. So let's start with the 3rd week lecture. And 3rd week have very interesting content. And I came with lots of who, why and when. And basically this lecture is about the probiotics. So let's start with the introductory part of probiotics.

Probiotics are the live micro-organisms that confer health benefits on the host while administered or taken in the adequate amount. And generally there are two species, that is lacto-bacillus and bifidobacterium. These two are grass species, generally recommended as safe; and mostly used as a probiotics. Now it is about the concept of the probiotics. Who introduced this term 'probiotics'. And its origin is very interesting. A Russian scientist, that is a Nobel Prize winner, Elie Metchnikoff, a good microbiologist.

He introduced this term ‘probiotics’, and he noticed that the people from Bulgaria, who are continuously taking the fermented foods; they have been reported with the good immunity and good gut micro-biota. So the term was introduced at the turn of the 20th century by the great scientist ElieMetchnikoff.

The word ‘probiotics’ originates, pro from Latin means for and bios is a Greek word, means life. In totality ‘for life’. And it was used in 1954 to indicate substances that were required for a healthy life-cycle. Now what are the probiotics. Which species we count. Which micro-organisms are probiotics. They are mainly two species. Lacto-bacillus species and Bifidobacterium species. And in Lacto-bacillus species, many strains are there, who are recommended as safe or grass strains like lacto-bacillus acidophilus, lactobacillus casei, lacto bacillus reuteri , lacto-bacillus bulgaricus, lacto-bacillus plantarum, and lacto-bacillus lactis also. In Bifidobacterium species; bifidobacerium bifidum, bifidobacterium longum, bifidobacterium infantis, bifidobacterium lactis and bifidobacterium adolescentis . These are the grass strains, and after so many tests have been conducted regarding their pathogenicity, safety. After that they have been declared as the safe strains, both lacto-bacillus and bifidobacterium species.

Now the ideal probiotic strains should have which kind of characteristics. To which strains we can say this is the ideal for using in the probiotic food. There are some characteristics or we can say, it should have some particular properties, like it should be known pathogenic or gras that is generally recommended as safe. Acid and bile tolerant and effective adhesion to the gut lining. This is a very essential characteristic we can say because if they adhere to the gut lining, only then they are able to multiply. And they will avoid the existence or interference of the pathogens that cause diseases. Then they should have the short generation time, and should able to survive the processing conditions like any kind of temperature, ph and other treatments. Then they should be anti-genotoxic property, genetically stable, and lactic acid producer. That all characteristics should be their in an ideal probiotic strain.

Then the probiotic attributes. Acid tolerance, bile tolerance and protoplast regeneration, anti-microbial activity. Because if they are able to produce some kind of anti-microbial, that inhibit the growth of the pathogens that is desired. Then gut colonization and lactose removal and the protease activity also. Then another question arises in everybody's mind when you will go through the content or learn about the probiotic. Who need probiotics? Everyone can take probiotics or some special kind of persons, they need probiotic. The answer is that anybody can have probiotic. They are the food. There is no need of the recommendation. But the people who are suffering from the diahrrea and weak immunity system, they need them. Otherwise anybody can have the probiotics. The people suffering from infection, weak immunity system and respiratory allergies; they can have probiotics. And persons suffering from inflammatory bowel disorders, constipation and intestinal infections. Then probiotics are very useful.

Then another why that is came in my mind that. What is the difference between probiotic and antibiotic. Very simple answer like, probiotics till now you can understand what are the probiotics. Probiotics are the live microbes. And when we take probiotics, they will interfere, they will inhibit the growth of the pathogens and support the growth of the good ones. But antibiotics, they don't know which one is the good, which one is the bad. When we'll take antibiotics, they will kill all types of the microbiota. And due to disturbance of the microbiota of the gastro-intestinal tract; it will get destroyed and then it leads to the that is diarrhea and other problems. Like in case of antibiotics it will destroy the microflora; all the microflora and then problem of digestion will take place.

But probiotics they promote the growth of the beneficial gut microflora, and thereby enhancing the digestion and also improve the immunity overall. Then some prerequisites of the probiotics. They should be non-pathogenic, non-toxic, resistant to gastric acid, because they have to survive in the stomach; till to reach the target site, they should be able to adhere the gut epithelial tissues; just to make colonization to interfere the entry of the pathogens, and they should be resistant to the bile side, and they should also be resistant anti-microbial mechanisms that operates in the gut.

Now stability or the viability of the probiotics is a very important factor. Like because viability is important because they are live microbes, they are not in the form of powder, any kind of concentrate, they are live microbes. So stability and viability is a big concern. If we don't maintain the particular temperature for their survival, their number will be decreased, and they will not provide us as much health benefit as required. So during fermentation, several factors affect the stability of the probiotics. Like composition of the medium. Composition of the medium is, it should have the nutrients enough for their survival like carbohydrates, proteins and fats. Only then they are able to survive during fermentation. Then physical conditions like temperature, pH and other factors. Then toxic bye-products and the presence of the oxygen also.

During downstream processing, extreme temperature conditions, oxygen stress and freezing and drying conditions are very important. Because if the temperature conditions or the temperature will be very extreme, then their ability will be affected. They are not able to survive at very high temperature. So these conditions affect the viability or stability of the probiotics. Then come to storage. After product development we come to the storage aspect. During storage, temperature is also important. Moisture conditions, oxygen stress and acidity of the carrier food, means in which food; it is a milk based food, a dairy based food or some known dairy food. What is the particular acidity of the food, that is important. And temperature; storage temperature is very very important here, because they are live microbes. If we will not maintain the low temperature, then thereby ability will be certainly affected. That is not desired. Then gastro-intestinal tract. In gastro-intestinal tract the acidity conditions of the stomach.Then enzymatic activity, composition of the environment and the amount of the secretion of the bile salt in the small intestine. These factors also affect the stability or the viability of the probiotics.

Then we should be aware like how these probiotics; we know that they are useful to us, we should take them. Then how they are useful? How they works? What happens when we have probiotics in our diet? What they did? This is called the mechanism of action of the probiotics. These probiotics works in many ways. Like many aspects are there. First aspect, that is very important is that they get colonization to the gut lining. Means when we take any probiotic food, the probiotic micro-organisms, they adhere to the membrane, and colonization will take place. And when pathogens will enter inside the body, they will not allow them to enter and inhibit their growth also. This is the one type of working or the mechanism of action of the probiotic. Secondly they compete with the nutrients also. They utilize the essential nutrients like proteins, fats which are required for the growth of the pathogens. And they exhaust and make that particular medium deficient, so that the pathogens, they pathogens can't be able to utilize the nutrients. Then another important mechanism of working of the probiotics is that they produce some kind of inhibitory substances like hydrogen peroxide and bacteriosis. The hydrogen peroxide, it is very harmful. It will not allow the growth of the ecoli. Ecoli is the pathogen. And similarly they produce such kind of inhibitory substances; bacteriocines and other acids, which inhibit the growth of pathogens.

Then another important aspect or mechanism of action of the probiotics is they degrade the toxic receptors. Production of inhibitory substances we have discussed. Then they have the antigenotoxic effects. Then antimutagenic effects. And certainly they will compete for the nutrients, and make them unavailable to the pathogens. So this is all about the probiotics. Where they originate? Who introduced the term 'probiotics', how they are useful, how they work? And who should take probiotics. And what is the difference between this probiotics and anti-biotics.

Thank You.

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Transcript

Hi! Welcome back. And this 2nd lecture of the 3rd week. Now what are prebiotics and who introduced this term 'prebiotics'. This term was introduced by Gibson and Roberfroid, who exchanged the word 'pro' from the probiotics with the 'pre' which means before or for. And the prebiotic, it should be a nondigestible food ingredient. Should have beneficial effects, selective stimulation and finally improve the host health. If we go for the definition actually of this prebiotics; these are the non-digestible, but fermentable oligo-saccharides, that are specially designed to change the composition and activity of the intestinal microbiota with prospect to promote the health of the host. These are the non-digestible. They don't contribute to the nutritive value of that food or food ingredient. They are non-digestible.

Then functions of prebiotics. What functions they perform?

- Prebiotics they maintain the optimal pH in the intestine, because pH is very important for the survival of the probiotics.
- And they stimulate immune system, because they enhance the growth or stimulate the growth of the probiotics.
- Then another function they perform, that they suppress the reproduction of the pathogens. If they suppress the reproduction of the pathogens, then they are not able to grow because they are harmful.
- They stimulate the peristalsis.
- They also reduce the formation of gases.
- And they stimulate the growth and reproduction of the useful microflora.
- As we have already discussed that these prebiotics are the food particularly for the probiotics.

Then what are the important criteria of prebiotic to say it is a good prebiotic. The classification criteria.

- First they should be resistant to upper gut tract.
- Then fermentation by the intestinal microbiota.
- Fermentation here, intestinal microbiota, it should be able to utilize the prebiotics.
- Then beneficial to the host health, definitely they will benefit if they are stimulating the growth of the probiotics.
- Then selective stimulation of the probiotics.
- Then stability to food processing treatments.

Food processing treatment here means any kind of heat treatment, cold treatment or any other kind of treatment. They should be stable to that kind of processing treatments. Then there are some commercial grade prebiotics. Food grade prebiotics that are available. That is Lactulose, FOS, that is fructo-oligosaccharide, galacto-oligosaccharide, isomalto-oligosaccharides, lacto-sucrose. Then xylooligosaccharides are also commercially available.

Then prebiotics, how they works, it's very important to know their mechanism of action, how they works, how they promote the growth. These prebiotics what they did. They enhanced the formation of short chain fatty acids and lactic acid as a fermentation product. Another way they also enhance the growth of the probiotic bacteria. That is the prerequisite, if we say a particular ingredient as a prebiotic. It should have as we say the quality or the property; that it will definitely enhance the growth of the probiotics. And another way of working is that they increase the level of calcium and magnesium in the colon.

Now another term that is correlated. Synbiotic. Synbiotic is a combo word of probiotic and prebiotic. And 'syn' means synergistic one. They support the growth of the probiotics. Then this term used. Synbiotic term is used when a product contain both probiotics and prebiotics. And the example is bifidobacteria and FOS. Here this example shows that if we are using bifidobacteria species in any probiotic product. Oligosaccharides that is fructo-oligosaccharides, this FOS in combination with this bifidobacteria, it will support or enhance or stimulate the growth of the bifidobacteria. This is proved one. We are not going to take any kind of prebiotic which will not able to support the growth of probiotics. That is not a desirable characteristic. So definitely this combo is a perfect combo.

Then synbiotics, we know that these are the both prebiotics and probiotics combination and they will improve the survival of the probiotics during the passage through the gastro-intestinal tract. They also provide the health advantage from live microbes, that they may not ensure from the prebiotics alone. They also stimulate probiotic growth by prebiotic components. Then the we should be aware about, their mechanism of action. How they works and how they are useful. The synbiotics they should have the synergistic effect, that we have discussed earlier also. Synergistic means they should be supportive. They should support the growth of the probiotics. And they should be stimulate the growth and survival of the probiotic. If synergy will be there, definitely they will stimulate the growth. Then they should be able to prevent the AOM-induced suppression of the natural killer- cells activity in Peyer's patches; that is a disease and in this they will express the production of natural killer-cells.

Then they should also be able to modify the composition of colonic bacteria ecosystem, which leads to alteration of the metabolic activity of the colon. Here you can see that prebiotics, probiotics and both things make synbiotics. Prebiotics in very simple language. It is very easy to remember also. Food for probiotic, probiotic live bacteria. These live bacteria need prebiotics as food. It's very simple. And probiotics, the substances that can only be metabolized by the gut bacteria and utilized by them and these probiotics are the active bacterial culture. We are aware about this. Synbiotics, they are combination of both pro and pre. That is very clear.

Now it is very important to mention the differences between probiotics and prebiotics. Very prominent differences we can see here that prebiotics are defined as non-living as compared to probiotics, nondigestible and special forms of fiber or carbohydrates. Three aspects are important; non-living, nondigestible and special form of fiber or carbohydrates. But in case of probiotics they are live microorganisms. That when taken by the host they will definitely confer the health benefits. Second important aspect regarding prebiotics is they are in the powder form, and they can survive the heat, cold and acid. But it doesn't happen in the case of probiotics. They are very fragile and very susceptible to the heat. They need a particular temperature for their maintenance to stay alive, to stay viable, and their number reduced over a period of time. Then third important difference, like prebiotics they perform their role, by nourishing the bacteria that live in the intestine. Very simple term, they nourish the probiotics. The probiotics they fight the harmful bacteria, that is called pathogens present in the gut and benefit the host.

So this is all about the differential features about the prebiotics, probiotics and synbiotics. And how they are useful to us. How they works, their mechanism of action and other parameters.

Thank You.

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Transcript

Hello welcome back. In the last lecture we have studied the different aspects of probiotics, prebiotics and synbiotics. Their mechanism of action and other aspects. And in this lecture we will discuss the Functional Probiotic Foods. Different functional probiotic dairy foods available are probiotic yoghurt, probiotic yoghurt cream, probiotic dahi, probiotic lassi, probiotic shrikhand, probiotic ice-cream, probiotic kulfi and probiotic butter. These are the dairy based probiotic products, that are available in the market.

Now this is a flowchart which shows the preparation of the probiotic dahi. How to prepare the probiotic dahi. And it is produced commercially. Here I just want to show you like how it is prepared. Like milk and skimmed milk. We take skimmed milk and milk to maintain the total solid of a milk. We add the skimmed milk, and then second step is clarification, just to remove the any foreign matter is there. Then heating. Heating will be done at 90°

temperature for 15 minutes. After that next step is cooling. And the milk is, will be cooled at 42° temperature. And the next step that is the important one, which will make it probiotic is the inoculation. We add the inoculation, that we say the starter culture of the probiotic, and the rate will be at 1%. And these probiotic cultures they are available at different places in which are maintained is their cultures, like National Dairy Research Institute, Karnal. CFTRI, Mysore, and other organizations also they have.

Then after addition of the starter culture, incubation will be there. The product will be incubated at 42° temperature, plus-minus 1 for 6-8 hours. And then the product will be there, that is probiotic dahi. The lowsheet is similar for dahi but the culture varies here. This is very important step. Then a very simple question that arises in my mind is, that why the dairy or milk based products, they plays the best role of a carrier for the probiotic strains or the probiotic micro-organisms. Lots of milk based fermented products are there, that I have mentioned earlier also; dahi, butter and the yoghurt and many more. What is the unique features of milk that we use it as a medium for the growth of the probiotics. There are many unique features of milk. It is a very complete nature's food, that contain almost all the nutrients in a very balanced amount for the growth and survival of the micro-organisms.

And second important aspect is that generally the dairy products we store at very low temperature. And this low temperature is desirable and it will help to maintain the viability of the culture, viability of the live microbes. That is a one very great advantage of the dairy products to be used as food matrix for the probiotics. Then another important feature is that the probiotics milk contain a very diverse range of the therapeutic components and some prebiotics also. Bio-active peptides, they are produced by the probiotic bacteria, and milk is enriched with prebiotic; this lactulose and galactooligo-saccharides. They are naturally present. We are not added them from the outside source. They are naturally present in the milk. So that milk has the very unique properties or very unique features. And used as a best probiotic vehicle.

Then they have the good buffering capacity also, which increase the, milk is a good buffer, which increase the survivability. Then fermentation of the milk with probiotic mechanisms. They increases the vitamin B12, folic acid, biotin, conjugated linoleic acid content also. Here the unique features of the dairy products as a food matrix for the probiotics has been discussed. In case of yoghurt or dahi, like it contains whey proteins. Whey proteins are there. And these whey proteins, they increase the viability of the probiotics. Viability of the probiotics, because they have the prebiotics also. And almost all the nutrients that are required for the growth of the probiotics. If we will see cheese, it is high pH, fat, solids, buffering capacity and low oxygen. All these things are desired for the growth of the probiotics.

Then come to the ice-cream that is a frozen product. And ice-cream has high pH near about 5.5 to 6.5. And this pH increases the survivability of lactic cultures during the storage. Then milk chocolates. It has higher total solids and lipids. And these total solids and lipids, they provide nutrients for the growth of the probiotics and they give protective effect to the probiotics, because they are providing them the food for their survival, so they provide the protective effect.

Now probiotics has so many advantages, we should know the availability and which are the players in the market who are producing or preparing these probiotic foods. Let's have a look.

- Nestle has launched Actiplus Probiotic Curd. It is available in the market.
- Mother Dairy, Nutrifit probiotic drink.
- Mother Dairy has also launched B-active probiotic curd.
- Yakult, probiotic drink is available.
- And Amul has launched probiotic ice-cream. Here Amul is the main market player. And Amul was first to introduce, this dahi, that is known as Prolife ice-cream. Profile ice-cream in 2007. And curd and lassi are also available of Amul company. Then flavored biotic yoghurt has also been launched by Amul. Then probiotic ice-cream and chocobar is also available.

- Then Heritage Foods, they have introduced their product, that is probiotic curd.

There are various institutes that are engaged in the probiotic research in India. And very famous, very popular one that is

- Central Food Technology & Research Institute. That is CFTRI, Mysore.
- NDRI, Karnal(Haryana). National Dairy Research Institute.
- Then Institute of Microbial Technology, Chandigarh.
- And NDDB that is, National Dairy Development Board, Anand(Gujarat)
- And Nestle Pvt Ltd, Panipat(Haryana), India.

So there are many research institutes who are very active in this preparation of this kind of products, as well as they are maintaining the cultures also for the probiotics and other related products.

Thank You.

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Transcript

Hello everyone. Welcome back. In the last lecture we have studied about the probiotics, synbiotics, prebiotics, their definition, parts and different organization and research institutes, who are busy in maintaining the cultures and other research activities related to probiotics and the various probiotic products available in the market.

Let's start with the lecture,' Beneficial Effects of Probiotics'. Probiotics have many benefits, like they

- improve tolerance of lactose, like people who are lactose intolerant. They improve the lactose intolerance
- Then protection from the gastro-enteritis.
- reduce the toxins because toxins are harmful or to the human beings.
- and cholesterol reduction.

- And probiotics they also help in the synthesis of vitamins also.
- And they are helpful to reduce the irritable bowel syndrome,
- improve the digestion and gut function.
- avoid the food allergy
- immune regulation
- And enhance the mineral bio-availability. Calcium, phosphorus and other minerals.

Now the physiological effects of probiotics. Here you can see that the probiotics help in the

- synthesis of the vitamins, like vitamin B12,
- and helps in the metabolism of the bile acids,
- production of short chain fatty acids; because short chain fatty acids they have the beneficial effects,
- reduction of the pH in the large bowel,
- immune system activation
- and development of mucosal barrier. They produce the barrier by colonization with the mucosal sites, and or inhibit the entry of the pathogens.

Here the probiotics, they have the anti-pathogenic activity. Anti-pathogenic activity. Like they work antimicrobial the during processing anti-microbial peptides and bacteriocins are produced. And these antimicrobial peptides and bacteriocins; they act as a inhibitory substances for the pathogens, means in the presence of these anti-microbial peptides and bactriocins pathogens, they are not able to grow. Then these anti-microbial peptides and bacteriocins; they are responsible or increase the permeability of the cytoplasmic membrane of the targeted cells. And when the permeability of the cytoplasmic membrane will increase, they release the small cytoplasmic particles, and by this depolarization of the membrane potential will be there, which leads to the; finally the cell death, and that is the aim. And all these activities are carried out when we are using probiotics. So probiotics act as a anti-pathogens.

Now the production of short chain fatty acids, how it takes place. Prebiotics like fructo-oligosaccharide or inulin also. We can take inulin also. And then the prebiotics, they are added to the probiotics; live microbes that is bifidobacteria or either it may be bifidobacteria species, it may be lacto-bacillus. When both are there, they cause fermentation, because this is a substrate and this is a micro-organism which utilize this substrate. During fermentation, what happened. Short chain fatty acids, they produce, that is called SCFA. And these short chain fatty acids they produce butyric acid, propionic acid and acetic acid, which inhibit the growth of the pathogens.

Probiotics have various health benefits and they help in lactose intolerance, the persons who are suffering from the lack or less secretion of the beta-galactosidase enzyme which is responsible for the lactose digestion for that persons. Probiotics also helpful in the thermogenesis and modification in the energy expenditure and that is related to the obesity also. And they protected or provide encapsulation of the flavors, act as a anti-diabetic. Another health benefit is immune-modulation, means any kind of change in the immune system that is desirable. Also reduced energy. Then rheological modification. Rheological modification means texture, flavor and other kinds of modifications that are desirable from the texture and flavor point of view of the product.

Now improvement of survival of probiotics. As we have discussed earlier that their survivability, their viability, it is very important. And there are some latest concept. By adopting that concept we can improve their viability. Because viability is a very important factor, we can say a very important aspect in case of probiotics. Like one example I should quote here, that is we use micro-encapsulation. Micro-encapsulation is a technique, that is a very latest technique. Now the nano-encapsulation is also there. And microencapsulation is very popular. Here what we do. We take a cell material or wall material and we encapsulate, if we want to encapsulate any kind of live micro-organisms or any kind of flavor compound we use the wall material, and that wall material provides protection to that parameter; it may be a flavor, it may be viability and other aspects.

And for the probiotics, heat from pelleting and length of storage, these are very important parameters. Like length of storage, because during storage, the viability definitely reduced because they are viable, they are live and it will cause the non-viability of the product. That is not desirable. But is natural, it happens during storage. So we have to improve the survivability and the technique is microencapsulation, I have told you in which we use some artificial biological membrane or any kind of covering that is based on the research finding. And provide good barrier property to that product.

And then prebiotics. Addition of prebiotics such as inulin or fructo-oligosaccharides etc, they will also helpful to improve the survivability of the probiotics. Then other food supplements like addition of food additives. In food additives we can say the fibers, the vitamins, the proteins which are lacking, because the growth will take place in a total medium which has, a media which has a protein source, energy source, carbohydrate source, vitamin source and other factors. So by micro-encapsulation, by use of prebiotics and by use of food supplements like fibers and vitamins, we can improve or we can enhance the survivability of the probiotics.

Then there are certainly some challenges to the Indian market or the Indian probiotic products, that are very important. First important aspect is misconception. Misconception is people are not aware of their beneficial effects. How they works, their availability. And they have misconception in mind that they are not good from the safety point of view, if we take them continuously. This is the one factor. Secondly, gap of knowledge. There is no proper advertisement about that product. So people are not aware which probiotics are available in the market. And another important aspect is the scientific evidence and clinical trials. Lots of researches have been done to prove them scientifically. They are beneficial and the clinical studies also. So there is lack of evidence. We have lots of research, lots of literature also, but still the clinical studies are less.

Then next challenge is the inappropriate labelling of the organism.

Here labelling part it is very important to mention the micro-organism, the probiotic culture that we are using. How much like the viable microbes are present at the time of packaging. How long we can use that probiotic product, because during storage definitely the number of the microbes will be lowest down. And one more aspect is safety studies. Like there are still the things are not clear regarding the safety issues like how much we should have how long we should have. So these terms should be very clear to make these products more reliable. Then access is limited to the affluent class only. The people's of the low economic group, they are not able to access them. Then technological challenges in maintaining the viability. And very prominent example of this technological challenges is to maintain the cold chain. In our Indian conditions it is very difficult to maintain the cold chain. And these microbes because they are live, I am telling again and again, they are live, they are viable. So temperature is very important parameter. If continuously we are not maintaining the temperature, their number will be reduced. And they will not able to provide the desired beneficial effect. So temperature is very important here.

Now this is a figure which gives you a complete idea, a complete, we can say a map about probiotics. Probiotics, they work importantly like in immuno-modulation, helps in normalized intestinal microbial composition and metabolic effects. Here we can see that immuno-modulation, it will leads to the balanced immune response, and control the, if the balance immune response will be there, it will be helpful to control the inflammatory bowel disease. And same parameter it will affect or alleviate food allergy or the symptoms of food allergy in infants and other peoples also. And immune-modulation also leads to strengthen the innate immunity. They are correlated. Then this normalized intestinal microbiota composition. If the microbiota composition or gut microbiota composition is healthy and normal, then it will control the irritable bowel syndrome. And this normalized intestinal microbiota, it will lead to the colonization resistance; means colonization resistance means they are able to make their colonies, make their growth. And their number will be dominate here, and when they their number will be dominated, their population will be dominated. They will not allow the pathogens to enter and even if some pathogens will enter, they will not allow them to survive even. And they will lead the suppression of the endogenous pathogens. Example, anti-biotic associated diarrhea.

Then it also suppress the endogenous pathogens. And example of this kind of infection is travelers diarrhea. Now come to the metabolic effects. In metabolic effects, they produce short chain fatty acids, folin synthesis of vitamins. For example folate and vitamin B12 to the colonic epithelium. And supply of short chain fatty acids, it will reduce risk of factors for the colon cancer. Then come to the this metabolic processes/effects. They help lower the level of toxigenic or mutagenic reactions in the gut. All these main three effects, they are correlated. The microbiota will be good in the gut. Then the colonization resistance will be there. And similarly you can see here the metabolic effects, it will lead to bile salt secretion and deconjugation and secretion also. And lactase hydrolysis. And if lactase hydrolysis will be there, it will improve the lactose intolerance. And the people who are lactose intolerant, it will help to them also.

So this figure gives a complete re-aspect of the probiotics, their working behavior and beneficial effects also. So this is all about the working pattern of the probiotics. The different probiotic products and all.

Thank You.

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Transcript

Hello welcome back. I hope you are enjoying the course with me. Let's start with the another lecture based on role of probiotics in disease prevention. Probiotics has many role in disease prevention. They protect from cancer, hepatic disease, helicobacter pylori infections, GIT infections. Also useful in pregnancy and anti-biotic associated diarrhea. Let's see what happened here with antibiotic associated diarrhea. Like if somebody have any disease, then the physician or doctor recommends antibiotic course, and after taking that antibiotic course, our gut microbiota got disturbed. And if the gut microbiota got disturbed then the infection chances will increase. The clostridium that is a pathogen. Their population will dominate and cause diarrhea. During antibiotic treatment gut microbiota it will get disturbed and population of the clostridium will be there and produces toxins, and it will leads to diarrhea.

Here you can see that if probiotics are there added in the diet, it will be helpful to maintain the microbiota in the gastro-intestinal tract. There is a case study regarding the traveler's diarrhea. And it is a clinical study which is a mixture of lactobacillus acidophilus and bifidobacterium bifidum taken to prevent the severity of the traveler's diarrhea. And it has been found that no adverse serious reaction to the probiotics were reported in these trials, when we are using bifidobacterium bifidum and lactobacillus acidophilus. And you can see that it is clinically also proved that these both microbes like bifido bacterium and lactobacillus they are useful.

Now we will discuss the role of probiotics in cancer prevention. How the probiotic helps to prevent the cancer. In the small intestine the enzymes that are present like glucosidase, beta-glucuronidase and azoreductase, nitroductase. These enzymes are responsible for the formation of the pre-carcinogens. It doesn't happen in only one day. Cancer doesn't take place in only one day. Like these enzymes they will lead to the production of the pre-carcinogens. Then if we introduce probiotics in the right amount, right doses; these pre-carcinogens are not able to convert to the active carcinogens. Means here the probiotics, they are stopping their conversion; of what, pre-carcinogens to active carcinogens. So probiotics are very important to prevent cancer.

And synbiotics that is a combination of oligo-fructose; oligo-fructose is a prebiotic you know and probiotic as lactobacillus acidophilus and lactobacillus casei . So 'syn', syn word we have used for synbiotics. This is pre and this is pro. And this is synbiotic supplementation in humans. It will help to decrease the level of gut flora enzymes. If the gut flora enzyme level will be reduced, then this process will not happen. In continuation probiotics also bind or inactivate the mutagenic compounds which cause mutations. And production of anti-mutagenic compounds suppress the growth of pre-carcinogenic bacteria which cause cancer and reduce the absorption of carcinogens, and enhance the immune function. Also influence on the bile salt concentration.

And by working on all these sides, in totality we can say, they prevent cancer growth or the cancer cells growth. In another case study for the colon cancer, conducted for the colon cancer, and the 80 people have been selected for this study; who had either colon cancer or that is benign polyps. They were randomly selected and lactobacillus rhamnosus and bifidobacterium longum, these are both probiotic strain. They have been taken and their effect has been studied on the tumor growth and intestines, and fair placebo was an inactive pill. Placebo means control. Whenever we are going to do any kind of clinical trial, we have one control also for the comparison. Means in this control we have not added any kind of probiotics. So what happened after 12 weeks, the patients who received the probiotic, showed decreased DNA damage in the lining of the colon and decreased growth of the production of the colon cells. The study has been conducted by Rafter et al. in 2007. And clinically trial has been conducted and it has been proved, and they have compared it with the control also. It happens only the patients who has been given the doses of the probiotics with these probiotic micro-organisms, that is lactobacillus rhamnosus and bifidobacterium longum.

Then role of probiotics in heart disease prevention. How heart disease can be prevented by use of the probiotics.

- They assimilate the cholesterol by the bacterial cells.
- Deconjugation of bile salts by bacterial acid hydrolyses.
- Cholestrol binding to the bacterial cell walls.
- And also reduction of hepatic cholesterol synthesis.
- And redistribution of cholesterol from plasma to liver.
- Then bacterial production of short chain fatty acids.

And earlier also we have seen that these short chain fatty acids are helpful to reduce the blood cholesterol level. So this is all about the role of the probiotics in the human beings, and in the animal studies also it is proved how useful they are, clinically even, and are able to prevent many kinds of serious diseases.

Thank You.

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Transcript

Hello, welcome back. By now we have studied the role of probiotics in prevention of various diseases like cancer, colon cancer, antibiotic associated diarrhea, regular diarrhea and other diseases. Let's continue with some more discussion about other serious diseases. This is a case study conducted in 2007 regarding the Crohn's disease in which 10 active Crohn's disease patients were given bifidobacterium and lactobacillus, continuously for 4 months. Bifidobacterium and lactobacillus as a probiotic. It was found that by the end of the therapy, 7 patients had improved clinical symptoms after combined probiotic and prebiotic therapy. And it has also been reported that 6 patients had complete response. So this case study shows the beneficial effect of bifidobacterium and lactobacillus on Crohn's disease.

Then in case of ulcerative colitis, the probiotic strain E.coli Nissle strain. It produce equally effective to Mesalazine which is an anti-inflammatory drug in maintaining remission of ulcerative colitis. Equal effect was reported. Then probiotics have shown efficacy to introduce and maintain remission in children, and in others with mild to moderate ulcerative colitis. So these probiotic strains they are equally important and equally effective as the drug suggested for ulcerative colitis. This is a case study in which probiotic strain lactobacillus acidophilus was reported and 187 patients were randomized to receive. There are 3 groups of patients, like 65 patients were given lactobacillus acidophilus , 60 patients were provided with Mesalazine alone that is the drug. And a third group, it was provided with lactobacillus acidophilus with Mesalazine, the probiotic with the drug; that has been given to the 62 patients.

Let's see the results. And it was reported by Zocco et al in 2006, that after 12 months the treatment with lactobacillus acidophilus was more effective than the standard treatment. Standard treatment means only if we are providing the drug alone with Mesalazine in prolonging the relapse-free time. So this is about the ulcerative colitis patients report.

Till now we have discussed the role of probiotics in prevention of many diseases. Have you heard that these probiotics are also helpful in the irritable bowel syndrome. Yes they also have the health benefits or they also works in irritable bowel syndrome. What happens? Probiotics reduce or cause reduction of the intestinal gas production, and also cause the modulation of the gut microbiota. Modulation here means if the gut microbiota are less in the colon, to it will modulate by various mechanisms, various pathways and enhance the gut microbiota. And by performing these functions, will benefit the irritable bowel syndrome. Another aspect of probiotics with helicobacter pylori infection. And here the probiotics helps to produce the anti-microbial substances.

Anti-microbial substances like hydrogen peroxide, bacteriosins and other acids which has a inhibitory effect on the pathogens like helicobacter. And side by side they stimulate the mucus secretion. And probiotics also compete for the adhesion sites, and here the probiotics compete for the adhesion sites. And they go for the colonization. And when colonization takes place, their population will dominate there. And if the probiotic microbes, their population will dominated. Then the pathogens if they enter inside the body by chance, they are not able to survive. Because they create a competitive environment for the survival of the pathogens. Then probiotics also stimulate specific and non-specific immune responses, and helpful in the prevention of helicobacter pylori infection.

Here is a case study. In this case study the effect of probiotic lactobacillus and bifidobacteria has been checked on the frequency and duration of the antibiotic associated diarrhea during helicobacter eradication. And this study was conducted by De Vrese et al ., in 2003. Here you can say this is days per subject related to diarrhea and this is the duration. And this is the placebo, that is the control one. And this is the treated one. Control here is that they are not provided with the lactobacillus or bifidobacterium or any kind of probiotic. And this shows that they have been provided with the probiotic. And here you can see the number of days are more approximate. 10 in case of studied person related to antibiotic associated diarrhea, but it is very less more than like less than 5 only in case of the persons provided with the probiotic. Similarly the duration was more antibiotic associated diarrhea that happens. The duration was more in case of control, who has not taken or who has not given any kind of probiotic. And here you can see that duration is very less. It is almost 2-3 days, as compared to the only we can say the 1or 2 days, as compared to the control.

Now the probiotic also plays important role in the lactose malabsorption. And the people who are suffering from lactose intolerance, they are, they can have the probiotics, and they are very useful. Lactose intolerance, we know that the persons they have very low levels or there is no secretion of the enzyme beta-galactosidase, that is responsible or that utilize the lactose, and lactose is a milk sugar. And the lactose intolerance person, they are not able to digest milk. But they can able to digest fermented products or probiotic products. We can say probiotic dahi, probiotic yoghurt. Why because in milk lactose is present. To they are not able to digest, but what happened during fermentation. This lactose is converted to lactic acid by the micro-organisms or whatever the culture we are using. So they are not able to tolerate lactose but they can take the fermented products. They can have the fermented products because in fermented products, lactose is converted to lactic acid.

Action of the bacterial beta-galactosidase on lactose, and it will give relief, provide relief from the lactose indigestion. And the strains; probiotic strains that has been reported are streptococcus thermophiles, lactobacillus delbrueckii subspecies bulgaricus. They improve the lactose digestion and reduce the symptoms of lactose intolerance. Here is a case study about the mal-absorption of the lactose. And this study is also conducted by De Vrese et al., in 2001. Here the effect has been studied with the fermented milk, with the live or heat killed lactobacilli. Two groups are here with the fermented milk and the heat killed lactobacilli on lactose mal-absorption by conducting this breath H2 test. And clinical symptoms have been recorded in 10 healthy persons, who are consuming this pasteurized milk and native fermented milk.

Let's see what happened. Here you can see that mal-digestion, in case of mal-digestion it shows the pasteurized product, they are taking the pasteurized product. Pasteurized means, here the lactobacilli or lactobacillus acidophilus; it has been killed. It is not live. And this one is the native milk or fermented one, in which we are using the lactobacilli. And you can see by this breath test, the graph shows that it is very high. And in case of probiotics it is very less comparatively. Then flatulence, the person who has taken this pasteurized milk. Pasteurized milk here there is no live microbes. They have been killed. They have almost higher lactadise, comparatively shows the higher range, as compared to the fermented or we can say the probiotics; provided with the probiotics.

Similar trend has been followed in case of pain and in case of diarrhea. The graph is high in the control. We can say the pasteurized product, the people who have taken pasteurized product and the fermented one or with the probiotic strain. And you can see in case of diarrhea, it is almost the, the person who has been provided with the this fermented milk. In case of diarrhea it is almost reported negligible. Then probiotics are also helpful or very beneficial in urogenital tract disorders. How it has contributed in the production of the anti-microbial substances. Whenever these probiotics are producing the anti-microbial substances, it may be H2O2, it may be bacteriosins, it may be some kind of acid. They always suppress the growth of the pathogens. And the purpose is only to suppress the growth or kill the pathogens, so that they can't survive.

Another important aspect is they compete for the adhesion sites. If they compete for the adhesion sites, their population will be comparatively lower than the pathogens, and there will be a competition for the adhesion; adhesion sites and if the adhesion sites are not available for the pathogens, then they are not able to grow. Then they compete with the pathogens with some other aspects also. And by causing all these factors, they relieve from the urogenital infections.

So we have studied the role of probiotics and case studies that has been clinically proved. That how important these probiotics are for our health and prevention of such kind of serious diseases.

Thank You.

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Transcript

Hello, welcome back. This is the last week of this course. I hope you have gone through the contents of 1st, 2nd and 3rd week. And in the last 3 weeks we have studied Functional Foods, Nutraceuticals; their definition, sources, categorization, probiotics as a functional foods, and significance of functional foods, prevention of diseases and promotion of health. So let's continue with this week. The first lecture. This is about functional foods with their health benefits. The functional components, they are distributed among the different plant foods. Like

- garlic, it contains sulphides and terpenes.
- green tea (phenolic acid, flavonoids and coumarins).
- soyabean, it contains phytates, phenolic acid, flavonoids and carotenoids.
- cereals contain phytates, phenolic acid, flavonoids.

- And citrus(phenolic acid, flavonoids, terpenes, carotenoids.
- And flax seeds (phenolic acid, flavonoids and lignans)

Now lots of functional foods are claiming for the different health benefits. Let's have a look about these health claims and their bio-active components part.

- The fortified margarines, they have plant sterols and plant stanol esters. And health benefit claim to reduce total and low density lipoprotein cholesterol and it is clinically proved. And recommended amount, how much should be taken. It is 1.3 gram per day. And if we talk about the regulatory status. This is Food and Drugs Administration approved health claim.
- Next is psyllium, a bio-active compound found in soluble fiber. It also reduce low density lipoprotein cholesterol. And the recommended dose is 1 gram per day. And this is also FDA approved. This is all are clinically proved.
- Next is soy. And the bio-active compound is protein. And it also claims the health benefit of the reduction of the LDL cholesterol. And recommended amount, how much we should take it. That is 25 gram per day. And this is also FDA approved.
- Then whole oat products which contain beta-glucan. They are also responsible for reducing the total and LDL cholesterol. And evidence proofs are very strong and clinically proved also. And the recommended dose is 3 gram per day. And this is also FDA approved.
- Then cranberry juice. And the bio-active compound present is proanthocyanidins. It reduces the urinary tract infections. Evidence are moderate, it is not clinically proved. Some trial has been done but it's not clearly chose the proven benefit of the cranberry juice. And the recommended amount or recommended dose is 300 ml per day. So it will fall under the category of conventional food not the FDA approved one.
- Then another type of functional foods with the health benefits. Garlic, you know that garlic contain the organo-sulphur

compounds, and health benefit is to reduce the LDL cholesterol.

- Similarly green tea contains catechins and health benefit is to reduce the risk of certain types of cancer. And this is the clinical trials but this is a survey based study. And the claims are weak to moderate. Not the very strong one. And it comes under the category of conventional foods.
- Then next functional food is spinach, kale and collard greens. And the bio-active compounds lutein and zeaxanthin. And it reduces the risk of age related muscular degeneration. And strength of evidence is weak to moderate. And the recommended amount is 6 grams per day. And it also comes under the category of conventional food or dietary supplements.
- Now tomato and processed tomato products. You know that tomato is a very rich source of lycopene. And one unique feature or unique parameter of lycopene is that it is the most active oxygen neutralizer, and also have a potent chemo-preventive activities. So lycopene is the most oxygen reducing bio-active compound. And it reduces the prostrate cancer. And the recommended doses are, we can have it daily. And it also comes under the category of conventional foods.
- Now another functional foods with the health benefits. That is animal based food, that is lamb, turkey, beef and dairy. And the bio-active compound is conjugated linoleic acid. Conjugated linoleic acid almost present in every food, but these dairy and animal based products, they have the higher quantities comparatively, and the foods related to the ruminants, they have comparatively the higher quantities. And the health benefit is to reduce breast cancer. And it has been the in-vivo and in-vitro studies has been conducted but the evidence are weak comparatively and it falls under the category of conventional foods.
- Then cruciferous vegetables like cabbage, cauliflower, broccoli. They have the glucosinolates, indoles. And they reduce risk of

certain types of cancer. And evidences here are also weak. And the recommended amount; recommended doses, 3 or more servings per week. And it is conventional food.

- Then come to the fermented dairy products. We can say probiotics. They support the gastrointestinal health and boost immunity. And trials has been conducted in-vivo as well as in-vitro. And the clinical data is comparatively less. And the recommended dose is we can have it daily. And it comes under conventional or dietary supplements.

Now during manufacturing or during preparation of foods, we are applying many processing, like heat treatment, any kind of chopping and blanching, pasteurization, sterilization and other kind of treatment. So definitely these treatments when applied on the food during processing. They will either enhance the bio active compounds, or otherwise they will also affect the amounts or the quantity of the bio-active compounds adversely. One example I would like to quote here is that these carotenoids and some lipid soluble food components, they shows the increased amounts during processing due to the increased surface area; because this chopping and cutting, they definitely increase the surface area. And the heat treatment also even able to breakdown the carbohydrate complex, and in case the bio-availability.

But in some cases it will also affect quantity or the amount of the bio-active compounds adversely. Another factor that will affect the processing of functional component is length of post-harvest storage. Means how long the product has been stored after harvesting. Then thermal processing, that I have told you; that is, it may be steam blanching, it may be pasteurization, it may be sterilization, it may be canning, and many other thermal treatments are there. It has been found that 30-80% of the bio-active isothiocyanates, they lost during heat processing. The amount is quite high, that is 30-80%. And high temperature more than 100° inactivates the key enzyme, that is myrosinase. And in case of garlic, if the heating of garlic at 60°-100° temperature results in significant losses of its anti-inflammatory, anti-cancer, anti-microbial and anti-oxidant activities.

So temperature here adversely affected the bio-availability or the bio-active compounds of the garlic. Then bio-availability of carotenoids have been shown to improve with the processing with cutting and chopping. Then brewing of tea leaves. Either it may be black tea or green tea. It was found that during brewing process it released 69-85% of the bio-active flavonoids. Within 3-5 minutes they leach out or dissolve in the hot water. And these studies have been conducted by Kim in 2000 about the effect of processing on functional components.

So this is all about the different types of bio-active compounds, different types of functional foods and the processing effects, how the heat treatment, chopping, cutting and the other factors affect the quality of these bio-active compounds.

Thank You.

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Transcript

Welcome back. Let us start with the 2nd lecture of this week. That is about the phytochemicals, phytosterols and dietary fibers. Now do you know what is the role of the phenolic compound to prevent the oxidative stress. I think you are aware that free radicals are harmful to us. And generally free radicals are the part of life. We can't avoid them. And I think you know that how much oxygen we need. A normal healthy human being consumes approximately 3.5 kg oxygen daily and the full oxygen amount can't be reduced. Then what happened. It will lead to the formation of free radicals and reactive oxygen species; that is ROS. And both are harmful.

So this reactive oxygen species, such as super-oxide, hydroxyl radical, peroxyl radical and alkoxy radical as well as hydrogen and other peroxides, they are harmful and phenolic compounds, they neutralize the effect and prevent the oxidative stress. Then come to phytochemicals. Phytochemicals are also important for the prevention, delaying the anti-carcinogenic effects. Let's see how phytochemicals have the

- anti-oxidant effect.
- Also affect the cell differentiation.
- Increase the activity of enzymes and helps in detoxification.
- Effect on DNA methylation.
- Maintenance of DNA repair.
- Increase the apoptosis of cancer cells, that is death of the cancer cells
- And decrease in cell proliferation.

So by creating all these factors, these phytochemicals helps in prevention or delay the occurrence of the cancer.

Now phytosterols. Phytosterols are also important plant equivalent of cholesterol in animals. And the most common bioactive phytosterols are beta-sitosterols, campesterol and stigmasterol. And the saturated derivatives of plant sterols are plant stanols such as sitostanols. Now come to the dietary fibers. We have discussed about dietary fibers earlier also. And dietary fibers importance is well known in our diet and proved also. The long fibrous structure of the dietary fibers, that is the long chain of carbohydrates; they allow to entrap the harmful toxins and carcinogens; the cancer causing particles in the digestive tract. And if the dietary fibers they entrap the harmful toxins and carcinogens, which further leads to the production of cancer. So here they are liberated, they are entrapped and thrown out and provide the anti-carcinogenic effects.

Then what is the role of polyphenols in low density lipoprotein cholesterol oxidation. These polyphenols, they have positive effect on prevention of low density lipoproteins, that is LDL cholesterol oxidation. It may be exerted by a free radical scavenging mechanism. And this will leadsto the oxidation of cholesterol. And it will cause the formation of plaque in the arteries. And due to the formation of the plaque in the arteries, there will be a blockage in the arteries. So polyphenols here plays very important role, and prevent the low density lipoprotein oxidation.

Now let us discuss a case study. This is about role of fish oils in rheumatoid arthritis. That is a very serious disease. What happened in rheumatoid arthritis? There is an inflammation in the joints and the patient suffers a lot, with the lot of pain and stiffness in the joints. Robert in 2005 conducted a clinical trial on the fish oil with the patients of rheumatoid arthritis and the group has been given fish oil supplementation 130 milligram per kg body weight. That is a fixed amount is mentioned 130 milligram per kg of the body weight each day. And what he reported that there was a decrease in number of stiff joints, duration of the morning stiffness, because morning stiffness is the main symptom we can say of this rheumatoid arthritis. The patients feel stiffness for a period of 30 or 45 minutes and movement becomes very difficult and pain also.

Now there is another case study. And in this case study the effect of special carbohydrate-protein bar. It is a bar and tomato juice. Their effects has been studied on the oxidative stress, and vascular endothelial dynamics in ultra marathon runners. Let's see what happened. This study was conducted and the bar was given, which have whey proteins and carbohydrates. Whey proteins and carbohydrates in a specific ratio; 1:1 ratio to the ultra marathon runners for a period of 2 months. And 16 peoples have been given either whey and 15 has been given tomato juice also, that is commercially available. We can see that administration of these 2 products to the runners for 2 month period, improve their oxidation state. While the tomato juice, both are improving like they as well as the tomato juice, they both are improving the oxidation state. But in addition to the oxidation state, the tomato juice also improve their vascular endothelial functions also.

So how important is phytochemicals, polyphenols and dietary fibers are in our life. That we have gone through in this lecture.

Thank You.

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Transcript

Hi, Welcome back. Let's start with a new lecture this of week. 'Role of Functional Foods in Management of Cancer and Cardio-Vascular Diseases'. Different sources, functions of this functional foods in cancer prevention. Here you can see that this is all about these all functional compounds and functional foods. They are related to cancer prevention. By taking all this we can prevent or delay the onset of the cancer. Like alpha-carotene, beta-carotene. And everyone knows that the dietary sources of alpha-carotene are yellow, orange and dark green leafy vegetables. Beta-carotene found in green leafy vegetables, and orange, yellow; fruits and vegetables. And what the effects. In moderate dose, it enhance gap junction intra-cellular communication, and give a particular benefit.

Then come to the lycopene. That is found in tomatoes, water melon, apricot and peaches. And lycopene, I have told you have a very unique feature. It is the most potent than alpha and beta carotene in inhibiting the cell growth of various human cancer cell lines. So lycopene plays a very important role in the prevention of cancer. Then lutein, that is found in dark green leafy vegetables. And lutein is very efficient in cell cycle progression and also inhibit the growth of the number of cancer cell types. Then flavonoids that is found in plants, synthesize in plants. And they also efficient in prevention as well as in treatment of many cancers.

Cardio-vascular diseases. You know that cardio-vascular diseases include heart diseases and stroke. And if we see the data, you can see that 60% of the deaths in economically developed countries occurs due to cardio-vascular diseases. The percentage is very high, that is 60%. And the reactive oxygen species, that is ROS and pathogenesis, both acute and chronic heart diseases, as a result of cumulative oxidative stress and to minimize hypertension. And the risk includes

- Attaining and maintaining a healthy body weight.
- Consuming a diet rich in calcium, phosphorus and magnesium.
- And consuming sodium in moderation. In moderate amount or less amount.

Then potential cardio-vascular protective effects of functional foods. These are the functional foods and these are the bio-active compounds. And the potential mechanism is by lowering the blood cholesterol. Because the blood cholesterol is directly related to the cardio-vascular problems

- In nuts, tocopherols, omega-3 fatty acids are present.
- Legumes have fiber and polyphenols.
- Fruits & vegetables(fiber)
- Margarine, it contains phytosterols.
- Fish oil is very well known for the omega-3 fatty acids.
- Fiber and phytochemicals are also found in whole grains.
- Daidzeins, they are the very important soya proteins.
- And flavonoids in dark chocolates.

Another potential mechanism; second is inhibition of the low density lipoprotein oxidation. First mechanism was lowering the blood cholesterol, and second is inhibition of the LDL oxidation.

- We can say that fish is rich in omega-3 fatty acids. Any fish like mackerel, tuna and other fishes.
- And tomato have lycopene.
- Green leafy vegetables and fruits with carotenoids.

So fish, tomato and green leafy vegetables and fruits. They have these bio-active compounds. And these bio-active compounds enable the oxidation of LDL. Here let's see the anti-cancer effects. How it happens? If anybody have the high animal fat protein diet. It will increase the susceptibility to colon cancer, by conversion of pre-carcinogens to carcinogens by intestinal microflora. And in intestinal microflora these enzymes are present, like glycosidase, beta-glucourinidase, azoreductase and nitroreductase. Here the production of the pre-carcinogens and these enzymes converts these pre-carcinogens to active carcinogens.

By now you are aware about the carcinogens. What are carcinogens? The cancer causing things, like so pre-carcinogens, they are converted to the active carcinogens. And here the onset will take place. It will start the problem of cancer. Here the use of that is probiotic strain, lactobacillus acidophilus and lactobacillus casei . This example we have taken from the micro-organism based probiotic foods. And use of these both strains of lactobacillus supplemented in humans helped to reduce the levels of these enzymes. If the level of these enzymes will be reduced. Then the secretion or the generation of these carcinogens will be less and it will act or plays important role in anti-cancer effects.

Now come to case study. In this the functional food and lifestyle approaches for the functional foods were used for the lifestyle approaches for diabetes prevention and management. The conclusion. It was concluded that the regular consumption of functional foods that may be associated with

- anti-oxygen properties
- anti-inflammatory properties
- insulin sensitivity
- and anti-cancer function.

They are considered as a very important part and helpful in prevention of type 2 diabetes mellitus. Another conclusion was the fruits, vegetables, oily fish, olive oil and tree nuts, they also serve as an important or model functional food; based on their natural content of nutraceuticals. And we know that the fruit, vegetable, fish oil as polyphenols, terpenoids, flavonoids, alkanoids, pigments and unsaturated fatty acids.

Then third important conclusion was polyphenol rich herbs. That is coffee, tea, green tea and black tea that we are taking. It has a very strong benefits on the metabolic and microvascular activities. Cholesterol and fasting glucose. It also lowers the fasting glucose level, and anti-inflammation, anti-oxidation in type 2 diabetes mellitus patients. And this is all about the importance of these bio-active compounds in prevention of any serious diseases like cancer and diabetes. And these bio-active compounds plays very important role. So we gain a very updated concept about the polyphenols and bio-active compounds.

Thank You

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Transcript

Hi! Welcome back. This is the last and final lecture of this week. That is about the role of functional foods in the management of Diabetes Mellitus and Parkinson Disease and other diseases. This is a research published in the journal 'Nutrients' by Manthou in 2014, about the glycemic response of carbohydrate protein bar with Ewe-Goat whey. Whey you know that is a byproduct of cheese and paneer industry. And in this case study, the glycemic index and glycemic load of functional food product. That is, Ewe-Goat Whey protein and carbohydrates in this bar. The whey protein and carbohydrates has been taken in equal ratio, that is 1:1. And the study has been conducted on 9 healthy volunteers, who randomly consumed the food and in two visits. Then after that what happened, the plasma glucose concentration was measured. The GI was determined by calculation of the incremental area under the curve. And glycemic index of the test food was found to be 5.18, while glycemic load of one test food serving was 1.09.

By seeing these results or these results indicate that the tested product can be classified as low GI. That is less than 55 and low glycemic load that is less than 10 of the food. It means this goat whey protein and carbohydrate enriched bar, it lowers the GI and GL. And the health benefits of low glycemic response due to whey protein consumption. That tested food could potentially promote health beyond basic nutrition. And this is that we want in case of functional foods. Because it promoted the health. It provided the basic nutrition, or promoted the health beyond the basic nutrition also by lowering the GI and GL of the food.

Heard about Parkinson disease. This is a neurodegenerative disease and occurs or appears in late in life. And this Parkinson disease, it may be triggered by some type of food products, if we study the role of the food products. Some food products increase the progression of the Parkinson disease like dairy products. And some food products may reduce the progression of Parkinson disease, like phytochemicals they have been found to decrease the Parkinson disease risk in the individual who are consuming the food enriched with carotene and beta-carotene. And this study was carried out by Miyake et al. in 2011. So in Parkinson disease, the diet also plays a very important role. There are some food which increase the progression of the disease. There are some foods which are responsible for the decrease of the progression of the disease. Increase was claimed by the dairy products. They caused the increase and decrease all the flavonoids, phytochemicals, phytosterols. They claiming for the decrease of the progression of the Parkinson disease.

Then DHA is also decreasing the effects. What is DHA here? It is the essential component of the phospholipid of cellular membrane in brain and retina of the eyes. DHA is very important for the development of the brain and vision and eye health. And FDA has approved and recommended its use in the full term grown infants. The use of both DHA and arachidonic acid. FDA has given approval for its daily use, that is this one, DHA and arachidonic acid. But there is some limited amount of the DHA. It should not exceed 2 grams per day from the food supplement. This is the amount. And soya that is genistein; soyabean isoflavone genistein. It is a source of protein and that also have a neuroprotective effect.

Here you can see that the food in the red promote neurodegeneration, and the food in green provide the neuroprotection. And this figure is about the role of nutrients in Parkinson disease. Here you can see that the red one towards milk side, they are the, they cause the neurodegeneration. And as we are moving towards the monounsaturated fatty acids, polyunsaturated fatty acids and vitamin C, D, E, riboflavin that is vitamin B2, carbohydrates and meat, it becomes protective. And here you can say these are the, cause the neuroprotection. That is omega-3, vegetables, fruits, carotenoids, genistein, tea, caffeine etc. So the dairy products, they cause neurodegeneration. And the this bio-active compounds or we can say the sources of the food products that contain bio-active compounds, they have the neuroprotective effects.

So this is very important role we have studied. The role of diet, the role of bioactive compounds in the prevention of the Parkinson disease. That is a very serious disease in the elderly people.

So with this we came to the end of this course. I congratulate all the students and participants, who have completed this course successfully. And I believe that they have gained the updated knowledge about the Functional Foods, Nutraceuticals, their sources. How the probiotics are important. Probiotics as a functional foods, and role of different bioactive compounds in the prevention and promotion of health. And various other aspects related to functional foods. I thank you all for participating in this course. Wish you a Good Luck.

Thank You Very Much.

This is where you can add appendices or other back matter.